Firm Responses to Clean Energy Tax Credit Transfers: Investment, Tax Planning, and Disclosures*

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Abstract: Clean energy tax credits are a primary policy tool for governments to encourage firms to invest in renewable energy. However, many clean energy firms are not tax-paying, rendering the credits unusable and potentially ineffective. The U.S. Inflation Reduction Act (IRA) of 2022 substantially changed the policy to permit firms to sell the credits for cash, providing immediate liquidity to the clean energy sector. This created a new multi-billion dollar market that significantly increased access to clean energy tax credits. Using a hand-collected dataset of disclosed tax credit transfers combined with financial and plant-level electric generator data, I examine how both selling (clean energy firms) and buying (unrelated companies) firms respond to this new market. On the seller side, I use difference-in-differences designs to examine investment responses and project quality. I find that selling firms increase capital expenditures and the number of electric generators after their first transfer, with the increase concentrated in clean technologies. On the buyer side, I show that participation in tax credit financing expands sharply and draws in new firms across many industries. Buying firms are more profitable and have higher ESG scores than other potential buyers. Finally, I show that buyers' effective tax rates decline following a purchase, and they incorporate these purchases into the environmental disclosures of their sustainability reports. Overall, my paper provides policy-relevant evidence that transferability shifts both real investment in clean energy and the set of firms financing green technologies.

Keywords: Firm responses, Clean energy tax credits, Inflation Reduction Act, Transferability, Investment, Sustainability reporting

JEL Codes: H25, H23, M41, M48, M14

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1 Introduction

In response to the growing economic and environmental consequences of climate change, governments have adopted increasingly ambitious emissions-reduction commitments and net-zero targets. Meeting these commitments requires large-scale private sector investment in clean energy, as the energy sector accounts for roughly three-quarters of global greenhouse gas emissions (International Energy Agency 2024a).[1] Such investment, however, is highly capital intensive, and the high upfront costs are consistently identified as a major barrier to the energy transition by academics, international organizations, and policymakers (Intergovernmental Panel on Climate Change 2023; International Energy Agency 2024b; De Haas et al. 2025; European Commission 2025). To lower these costs and encourage firms to invest, governments in recent years increasingly rely on clean energy tax credits, which reduce the taxes a firm owes when it invests in or produces clean energy. In this paper, I study how firms respond to clean energy tax credits.

These tax credits, however, only provide an immediate benefit to firms that pay enough taxes to use them. Given the high costs of clean energy projects, many clean energy firms barely break even and often pay little or no tax during project development. Therefore, they cannot use the tax credits in the short run, substantially weakening the investment incentive and undermining the policy's goal. Before 2023, a firm in this position had only one way to turn its credits into cash, a relatively complex arrangement known as tax-equity financing that in practice was used by a limited number of companies (Garrett and Shive 2025).[2]

The U.S. Inflation Reduction Act (IRA) of 2022 dramatically changed this by making clean energy tax credits transferable. A clean energy firm can now sell credits to an unrelated firm in exchange for immediate cash rather than waiting until it becomes tax-paying. The buyer acquires the credit at a discount but applies its full value against its own taxes. By removing the need for a tax-equity partnership and creating a new tax credit market, transferability is intended to expand participation beyond the large institutions that historically dominated the tax-equity market.

This change represents a major shift. For example, former Treasury Secretary Janet L. Yellen described the new monetization tool as a "force multiplier, enabling companies to realize far greater value" from clean energy incentives. Lael Brainard, then White House

[1] Estimated annual global investment needs range from $3.5 to 9.2 trillion (World Economic Forum 2023).

[2] For example, in congressional testimony, Craig Gordon of Invenergy, a leading U.S. clean energy firm, emphasized that tax equity for renewable projects had always been "limited and highly concentrated" (U.S. House 2021a). Separately, John Larsen of Rhodium Group, an independent research organization, described the market as having "hit its functional limits" and reported that firms could not obtain financing or faced very high financing costs (U.S. House 2021b).

National Economic Advisor, stated that transferability lets "more small businesses, startups, and other businesses [...] benefit from game-changing clean energy tax credits" (U.S. Department of the Treasury 2024). Reflecting this substantial shift, transfer volume reached roughly $24 billion in 2024, surpassing the size of the tax-equity market ($20 billion in annual financing) in only its second year. In the next year, the volume nearly *doubled* to $42 billion (Crux 2025).[3] My data on the number of firms participating in this new market confirm this substantial increase (see Figure 1).

Despite the scale and growth of this market, little is known about which firms buy and sell tax credits and whether transferability has resulted in greater clean energy investment and financing. In this paper, I examine the responses of both the firms that generate and sell tax credits (*sellers*), as well as the firms that buy them (*buyers*). Prior work on clean energy tax incentives primarily focuses on the firms and projects receiving the incentive or financing (Garrett and Shive 2025; Chan and Fischer 2026). Studying sellers allows me to assess whether transferability increases clean energy investment, while studying buyers reveals which firms supply funds and what motivates their participation. Together, these analyses provide a holistic view on the policy's impact. This is especially important because the IRA represents the largest climate investment in U.S. history, with most of that support delivered through tax incentives (Tax Foundation 2022). Not only does this paper evaluate whether the incentives motivate clean energy investment but it also informs the ongoing policy debate over the future of clean energy tax incentives. Specifically, the One Big Beautiful Bill Act of 2025 accelerated the phase-out of several credits, but lawmakers from both parties have since introduced bills to restore them.[4]

I start by focusing on the clean energy firms who generate and *sell* the tax credits. I test two hypotheses; whether tax credit transferability (i) increases clean energy investment and, (ii) whether it affects the quality of that investment. I begin with investment levels. Theory suggests that transferability will increase investment for two reasons (Hall and Jorgenson 1967). The first is a *monetization* channel. Firms that pay little or no tax cannot fully realize the value of the credits they generate. Allowing these firms to sell their credits for cash raises the realized value of the tax incentive, lowers the after-tax cost of eligible investment, and can make additional projects profitable, thereby increasing investment. The second is a *financing* channel. The cash proceeds increase internal funds and reduce reliance on external financing, which can enable firms to undertake projects they might otherwise defer or forgo, particularly when they are financially constrained (De Haas et al.

[3]By further comparison, the market for transferable credits now approaches the size of the entire federal research and development credit, which totaled roughly $43 billion in 2022 (Internal Revenue Service 2025).

[4]See the Energy Bills Relief Act, introduced in the House by Democrats in March 2026, and the American Energy Dominance Act, introduced in the House by Republicans in April 2026.

2025; Blonz et al. 2026). However, while theory suggests we should see an increase, the magnitude of the effect remains an empirical question. Tax-equity financing already allowed some firms to monetize their credits. To the extent this market was sufficient, there would be a limited impact of the IRA (Garrett and Shive 2025). Transaction costs, buyer demand, and uncertainty in the new transfer market may further reduce the value firms realize and limit their investment response (Gallemore et al. 2025).

My second hypothesis concerns the quality of investment. Transferability increases the realized credit value and the internal funds available to develop, complete, and operate generating assets, which can improve the the performance of new projects and existing assets. However, transferability may also reduce investment quality. First, by lowering the after-tax cost of investment, more marginal projects, meaning those with lower expected operating productivity, may be profitable (Eichfelder et al. 2023; Chan and Fischer 2026). Second, new buyers may be less able to monitor the projects (Garrett and Shive 2025). Overall, the effect of transferability on investment quality is ambiguous ex ante.

To study firms' responses to tax credit transferability, I hand-collect a novel dataset of clean energy tax credit transfers. Because no data source identifies these transactions, I assemble the data from SEC filings and public announcements and identify 218 firms that participate as buyers or sellers between 2023 and 2025. For each transfer, I record the disclosed parties and, when available, the credit value and transaction price.[5] To my knowledge, this is the first systematic firm-level dataset to identify participants on both sides of the transferable tax credit market.

I begin the seller-side analysis by testing my first hypothesis about the level of investment. I link the hand-collected sample of clean energy firms that sell tax credits to financial statement data from Compustat and measure investment using firms' capital expenditures. To complement the financial statement analysis, I use more granular plant-level data from the U.S. Energy Information Administration (EIA). These data allow me to observe a firm's number of electric generators, which are the predominant asset purchased by clean energy firms. I use difference-in-differences (DiD) designs to compare investment around firms' first tax credit transfer with investment among otherwise eligible firms that do not transfer credits.

My main finding is that firms increase investment after entering the transfer market. In my preferred specification, capital expenditures increase by 1.2 percentage points of lagged total assets. This is equivalent to an increase of roughly 17 percent relative to the

[5] I observe a transfer only when at least one party publicly discloses it. I benchmark the aggregate volume identified in my sample against external estimates of the transfer market and confirm that the sample captures a substantial share of estimated market volume over the sample period.

pre-treatment mean and translates to approximately $374 million in additional annual capital spending per treated firm. The EIA data show that firms add approximately 9 generators. I then classify generators as clean or non-clean and find that the increase is concentrated in clean generators, including solar, wind, and battery storage. There is no corresponding change in non-clean generators, such as coal and gas. These findings suggest that transferability increases investment in the clean technologies targeted by the policy.

A concern inherent to my setting is that firms choose whether to sell a tax credit and may thus differ from control firms in ways that also affect investment. I take three steps to address these concerns and improve the comparability between treated and control firms. First, I restrict the sample to clean energy firms eligible to generate tax credits. Second, I use entropy balancing to align treated and control firms on observable characteristics. Third, I include firm and year fixed effects to absorb time-invariant firm differences and shocks common to firms in a given year. Finally, my event-study plots confirm similar pre-treatment trends in investment for treated and control firms, supporting the parallel trends assumption.

I next use cross-sectional tests to examine the mechanism driving the investment response. I find evidence consistent with both the monetization and financing channel contributing to the investment response. The monetization channel should matter more for firms that pay little or no tax, since they have the greatest difficulty using the credits they generate. Consistent with this, I find that the increase in investment is stronger among firms with larger pretax losses, net operating losses, and lower profitability. The financing channel should matter more for firms with limited access to external finance. Across several measures of financing constraints, I find a stronger investment response among more constrained firms.

I then turn to my second hypothesis for the sellers and examine whether the additional investment is accompanied by changes in investment quality. An additional benefit of the EIA data is that they allow me to measure investment quality using a generator's realized operating performance. My main measure is the capacity factor, which captures how much output a generator produces relative to its potential capacity, where a higher ratio signals greater investment quality. I conduct two tests. First, I examine the performance of generators added after the first transfer to assess whether transferability draws in lower-quality marginal projects. Second, I study the performance of generators already in operation to assess whether proceeds from credit sales improve the operation of existing assets. Overall, I find little effect, and additional tests confirm the null result. Nonetheless, these null results are informative. First, this means the investment added after a transfer does not perform worse than firms' existing assets, suggesting that transferability does not

pull firms toward lower-quality projects. Second, when viewed with the investment result it suggests that the tax credit proceeds are spent on expansion rather than improving existing assets. This is important because increasing clean energy capacity was a key policy goal of the IRA.

In the second part of the paper, I provide descriptive evidence on clean energy tax credit *buyers* and the determinants of their market participation. I first examine whether transferability broadened participation beyond traditional tax-equity investors. Prior to the IRA, the number of new tax-equity partnerships ranges from 5 to 12 per year. After the IRA, the number of firms purchasing tax credits exceeds 50 in 2024 and reaches roughly 100 in 2025. These new buyers are economically significant, with publicly traded buyers having an average market capitalization of $25.4 billion. Participation also extends across a wider range of industries. Earlier tax-equity investors were concentrated almost entirely in finance and insurance, whereas tax credit buyers span 13 additional industries, including manufacturing, retail, and accommodation and food service. Examples of tax credit buyers include Keurig Dr Pepper, Tripadvisor, Best Buy Co., and McDonald's Corporation. To examine what predicts entry into this market, I conduct a determinants analysis and find that firm size and high tax payments predict participation. These results suggest that the primary driver of participation for a firm is to reduce its tax liability.

I next examine the effects of purchasing tax credits on buyers. I begin by estimating the effect on buyers' effective tax rates (ETRs). I first quantify the tax benefit of credit purchases using information disclosed in buyers' 10-K tax footnotes. For buyers with sufficient information, I find an average benefit equivalent to a 1.2 percentage-point reduction in ETR. This reduction is economically meaningful and comparable to international tax planning strategies (e.g., Dyreng and Lindsey 2009). I then examine whether buyers' overall ETRs decline around their first purchase using a stacked DiD design. After balancing buyers and comparison firms on observable characteristics, I find a statistically significant decrease in GAAP ETR in the year the credit is first recognized, consistent with these credits delivering value relatively quickly to these buyers.

Beyond tax savings, I also study how firms disclose credit purchases in their sustainability reports to examine whether firms also benefit in a second, disclosure-related manner, from these credits. Specifically, a tax credit purchase can link a buyer to clean energy investment and provide content that can be incorporated into broader sustainability narratives (Lin et al. 2024). Because purchasing a credit does not require ownership or operation of the underlying project, however, the connection between the purchase and the project's environmental benefits, such as emissions reductions, is indirect. This creates scope for firms to present tax credit purchases as evidence of environmental commitment

even when the transaction has a limited connection to the firm's own environmental impact, raising potential greenwashing concerns (Heath et al. 2023; Bilicka et al. 2025; Duchin et al. 2025). To examine how firms present these transactions, I collect and review buyers' sustainability reports. My exploratory review shows that approximately half of the buyers that publish sustainability reports describe their tax credit purchases as environmental investments or as evidence of their broader environmental commitments.[6]

This paper makes several contributions. First, I contribute to the literature on the real effects of environmental tax policy (Lester and Olbert 2025). One strand of this literature studies carbon taxes and their effects on capital investment (Jacob and Zerwer 2024), R&D spending (Brown et al. 2022), firm emissions (Martinsson et al. 2024; Colmer et al. 2025), and carbon leakage (Känzig et al. 2024). A related, but nascent, strand of concurrent work examines green investment incentives, including the effects of investment tax credits on electric utilities' generation and emissions (Chan and Fischer 2026) and the effects of government loans and subsidies on corporate environmental performance (Cincotta and Koester 2026). I extend this literature by studying transferability as a distinct feature of environmental tax policy design. Transferability changes firms' ability to realize the value of an existing tax incentive rather than its availability or amount. My findings show that the effectiveness of clean energy tax incentives depends not only on their value on paper but also on firms' ability to monetize them and finance the underlying clean energy investment.

Second, I provide some of the first systematic evidence on the corporate buyers of clean energy tax credits. Prior work on clean energy tax incentives and tax-equity financing primarily examines the firms and projects receiving the incentive or financing (Garrett and Shive 2025; Blonz et al. 2026; Chan and Fischer 2026). I complement this literature by identifying which firms purchase tax credits and what they gain from participating in the market. I find that buyer participation expands substantially relative to the pre-IRA tax-equity market and extends beyond the financial institutions that traditionally supplied tax-equity capital. I further show that buyers realize two benefits, a direct tax benefit from purchasing credits, reflected in lower effective tax rates, and disclosures in their sustainability reports. This evidence also contributes to the clean energy financing literature by documenting how a broader corporate buyer base supports the monetization of tax incentives and the financing of clean energy projects (De Haas et al. 2025; Lanteri and Rampini 2025).

Finally, the paper provides evidence relevant to the ongoing policy debate over clean energy tax incentives. The market for transferable credits has grown rapidly as

[6]For example, one corporate buyer describes its purchase of renewable energy tax credits in its CSR report as a way "to further bolster our own commitment to the energy transition."

policymakers reconsider the role of the underlying credits in encouraging clean energy investment. My findings show that transferability increases sellers' clean energy investment while broadening corporate participation in clean energy financing beyond traditional tax-equity investors. Together, these results inform the broader debate over the effectiveness and how these incentives should be structured to reach firms and encourage investment.

2 The New Market for Tax Credits

Clean energy tax credits reward firms for investing in or producing clean energy. In the U.S., the two principal credits are the Investment Tax Credit under Sections 48 and 48E, which is based on a project's eligible investment cost and earned when the project is placed in service, and the Production Tax Credit under Sections 45 and 45Y, which is based on electricity generated and earned over the years a project operates.[7] The Inflation Reduction Act of 2022 expanded existing clean energy credits and introduced new incentives for technologies such as clean hydrogen, advanced manufacturing, carbon capture, and clean fuels.[8] For taxable firms, these credits reduce federal income tax liability dollar-for-dollar, thereby lowering the after-tax cost of clean energy investment. Figure 2 Panel (a) illustrates this mechanism for a clean energy firm. The firm earns a tax credit by undertaking an eligible investment or producing eligible clean energy and claims the credit against its federal tax liability. The credit reduces the amount of tax the firm owes by its face value. The value of the incentive to the firm, however, depends on whether it owes enough taxes to use it. Firms that owe little or no tax cannot use the full credit immediately.

This is a common problem for clean energy firms. Many are young and face large upfront investment costs, so they owe little or no tax in the years they generate credits. Before the IRA, the primary way for firms to turn credits they could not use themselves into cash was tax-equity financing. Figure 2 Panel (b) illustrates a typical tax-equity transaction. In this structure, a clean energy firm partners with an investor that can use the credits to reduce its own taxes. The investor provides funding in exchange for a share of project ownership, through which it receives the project's tax credits and other tax benefits. These arrangements are complex and costly and require substantial legal, tax, and accounting expertise. Tax-equity investment has therefore historically been concentrated among a relatively small group of investors, particularly banks, insurers, and large corporations, limiting access for many clean energy firms. In addition, the legal, administrative, and

[7] For facilities placed in service after 2024, Sections 48E and 45Y replace Sections 48 and 45. The transition rules allow certain projects that began construction before 2025 to remain eligible for Sections 48 and 45.
[8] Appendix Table A.1 lists the eligible credits, their statutory sections, and the technologies they cover.

compliance costs associated with tax-equity structures can reduce the value firms realize from the underlying tax benefits by an estimated 20 to 30 percent (U.S. Government Accountability Office 2015; Congressional Research Service 2019).[9]

The IRA introduced transferability under Section 6418 of the Internal Revenue Code as an alternative.[10] Figure 2 Panel (c) illustrates this mechanism. A clean energy firm that earns an eligible credit can sell all or part of it to an unrelated buyer in exchange for cash.[11] The buyer uses the credit to reduce its own tax payments. For applicable credits, a buyer that cannot use the full credit in the year of purchase can carry the unused portion back up to three years or forward up to twenty years.[12] The price for the tax credit is below the face value. The seller therefore accepts a discount to receive cash, while the buyer saves more in taxes than it pays for the credit. In 2024, production tax credits traded at an average of 95 cents per dollar of credit and investment tax credits at 92.5 cents (Crux 2025), allowing sellers to realize substantially more of the credit's face value than under traditional tax-equity financing. The discount reflects transaction costs, tax risk, and market conditions. Unlike tax-equity financing, a direct transfer does not require the buyer to own a share and fund the underlying project. Although transfers do not require an intermediary, many are arranged through platforms and brokers that match buyers and sellers and help complete the transaction.[13]

Two features of transferability are notable. First, the credit can be separated from ownership of the underlying project. The buyer has no operational or governance rights and does not receive an ownership interest or claim on the project's operating cash flows. Second, the buyer nonetheless bears the tax risk associated with the credit. For credits subject to recapture, the buyer is liable for any recapture attributable to the portion it purchased, even though it does not control the project. For example, an investment tax

[9] Tax-equity transactions also monetize accelerated depreciation in addition to the credits, a benefit transferability does not provide. Hybrid structures that combine a tax-equity partnership with a transfer have become common.

[10] Policymakers favored transferability over direct subsidies or expanded direct-payment programs for several reasons. The IRA passed through budget reconciliation, which limited new direct spending, and a transferable credit is scored as forgone revenue rather than government spending, thereby preserving the political framing of the support as a tax provision. Transferability also shifts the task of evaluating and pricing project risk to private market participants rather than the government, which is intended to limit fraud and improve project selection.

[11] The seller must register the property that generates the credit with the IRS and obtain a registration number. The seller and buyer then complete a transfer election statement, which each attaches to its tax return. The statement must be completed before either party files its relevant return.

[12] The credit remains subject to the general business credit limitation under Section 38. Any unused portion of an applicable credit can be carried back up to three years, beginning with the earliest year, and forward up to twenty years.

[13] Platforms such as Crux and Reunion match counterparties and assist with pricing, due diligence, documentation, and compliance.

credit can be recaptured during the five years after the project is placed in service if the underlying property is sold or otherwise ceases to qualify, including when financial distress or insolvency results in a disposition of the project.[14]

The IRA also restructured the amount of the credit available for transfer. For many clean energy credits, firms qualify for a base credit and can receive a larger amount by meeting additional requirements. Meeting prevailing-wage and apprenticeship requirements increases the base credit amount fivefold.[15] Certain credits also provide additional bonuses for meeting domestic content requirements or locating projects in qualifying energy or low-income communities. As a result, the dollar value of an earned credit can vary across otherwise similar projects. These provisions affect the amount of the credit generated and therefore the amount available for transfer, but not the transfer mechanism itself. Once the eligible credit amount is determined, including any applicable bonus amounts, the seller can transfer all or part of it.

The One Big Beautiful Bill Act of 2025 accelerated the termination of several clean energy credits, shortening the period during which new projects can qualify. These changes were enacted in July 2025, during the final year of my sample, and primarily affect the eligibility of future projects rather than the transfer mechanism that I study.

3 Hypothesis Development Seller-Side

I first ask whether transferability affects a firm's clean energy investment. In the neoclassical user cost of capital framework of Hall and Jorgenson (1967), an investment credit of k per dollar of investment reduces the price of new capital, lowering the user cost of capital and raising a firm's desired capital stock.[16] A tax credit therefore lowers the after-tax cost of investing in eligible assets and should encourage firms to invest more. The framework rests on two assumptions. Firms realize the full value of k, and capital markets are perfect, so that internal and external funds are perfect substitutes and a firm's cost of funds r does not depend on its access to internal finance. In practice, two frictions prevent these assumptions from holding. The first is a monetization friction. Firms that owe little or no tax may be unable to use their credits in the short run, so the effective value

[14] The seller is responsible for substantiating the underlying credit and must notify the buyer of a recapture event. Buyers can obtain insurance against recapture risk to reduce financial losses.

[15] For example, the Section 48E investment credit increases from 6 percent to 30 percent of qualified investment when these requirements are satisfied.

[16] A standard representation of the user cost of capital is $c = q(r+\delta)\frac{1-k-uz}{1-u}$, where q is the price of capital, r is the cost of funds, δ is the depreciation rate, u is the corporate tax rate, z is the present value of depreciation deductions per dollar of investment, and k is the tax credit per dollar of investment.

of the credit falls below k. The second is a financing friction. Clean energy projects are capital intensive, and financially constrained firms may face costly external financing that raises their effective cost of funds, r. Transferability can address these frictions through two channels.

The monetization channel raises the value firms realize from credits they cannot use themselves.[17] Firms with access to tax-equity financing can monetize the credit indirectly, but these arrangements are costly to structure and reduce the value ultimately realized. The effective value of the credit can therefore fall below its statutory amount k, so that projects that would clear the investment hurdle at the credit's full value no longer do. Transferability addresses this friction through a monetization channel. By allowing firms to sell credits for cash, it can raise the realized value of the credit closer to k, lowering the user cost of capital and making additional clean energy projects profitable. This channel should matter most for firms that have difficulty using the credits themselves, so I expect a stronger investment response among firms with low or negative pretax income.

The financing channel provides funds that can support additional investment. Clean energy generation is capital intensive, with a large share of project costs incurred before projects begin generating revenue (Blonz et al. 2026). Whether a project proceeds therefore often depends on the firm's ability to raise external financing, even when the project has positive net present value. For financially constrained firms, costly external financing can raise the effective cost of funds, r, and prevent otherwise profitable investment. Selling a credit provides cash that can be reinvested in existing or new projects. These proceeds increase internal funds and reduce the external financing the firm must raise (De Haas et al. 2025), which can lower its effective cost of funds and therefore r. Consistent with this, Blonz et al. (2026) find that smaller firms experience a decline in borrowing costs relative to larger firms following the IRA and argue that these firms are more likely to benefit from transferability. By lowering the user cost of capital, this channel can also increase clean energy investment, in particular among financially constrained firms.

These two channels can operate together. A firm that owes little or no tax may also face greater financing constraints (Blonz et al. 2026). For such a firm, transferability can both increase the value it receives from its credits and improve its ability to fund investment. The cross-sectional predictions therefore describe where each channel should matter most, while allowing both to contribute to the same firm's investment response.

[17] Unused credits can generally be carried to other tax years. Carrying credits forward delays the tax benefit and reduces its present value, particularly for firms that may take several years to earn enough profits to use them.

Although theory predicts an increase in investment, the size of the response depends partly on how much transferability improves on existing monetization and financing options (Garrett and Shive 2025). Firms with access to tax equity could monetize their credits and obtain project financing before the reform, although these arrangements were relatively costly and complex. Transferability may therefore represent a larger change for firms that previously lacked access to tax equity, while also offering existing tax-equity users a simpler and potentially less costly way to monetize their credits.

The investment response also depends on how effectively the newly created transfer market works. Ex ante it is unclear whether buyers and sellers participate and whether transactions can be completed at a reasonable cost. Finding a buyer, pricing the credit, conducting due diligence, and completing the transaction impose costs that reduce the net value realized by the seller. The market also requires sufficient demand from buyers that owe enough tax to use the credits. If transaction costs are high or buyer demand is limited, transferability may provide only a modest improvement over existing monetization options. Uncertainty about the new market and the durability of the underlying tax incentives may further limit firms' responses (Gallemore et al. 2025). Although these factors may limit the investment response, both the monetization and financing channels predict a positive effect. I therefore predict that transferability increases clean energy investment.

H1: *Tax credit transferability increases the level of clean energy investment.*

The preceding discussion considers whether transferability increases the amount firms invest. An increase in investment, however, does not necessarily imply an increase in investment quality, which I define as the realized operating performance of a firm's investments. Transferability may change both the projects firms undertake and the resources available to develop and operate them. I therefore next ask whether transferability affects the quality of clean energy investment.

On the one hand, transferability can improve investment quality by increasing the value firms realize from their credits and the funds available for investment. The resulting reduction in the after-tax cost of investment can make higher-quality equipment more affordable, consistent with evidence that tax incentives can affect the quality of capital goods firms purchase (Goolsbee 2004). Credit sales also provide cash that firms can use to develop, complete, and operate generating assets. A financially constrained firm may otherwise lack the resources to fully fund project development, purchase suitable equipment, or maintain existing assets. Additional funds can therefore improve the performance of newly added generators as well as generators already in operation.

On the other hand, transferability can reduce the average quality of investment for two reasons. First, by increasing the value firms realize from tax credits and lowering the effective cost of capital, transferability makes additional projects profitable. These marginal projects may have lower expected operating productivity than the projects firms would have undertaken absent transferability.[18] Thus, transferability can change not only how much firms invest but also the composition of projects that firms undertake. Consistent with this possibility, Eichfelder et al. (2023) study bonus depreciation incentives for business investment in East Germany and find that the tax incentive increases investment while generating smaller increases in future revenues. Similarly, Chan and Fischer (2026) find that U.S. electric utilities claiming investment tax credits increase investment but experience smaller increases in subsequent electricity generation. These findings suggest that tax incentives can increase investment without producing a corresponding increase in operating performance. Transferability may therefore increase investment while lowering the average performance of the projects undertaken.

Second, transferability may weaken project monitoring to the extent that it substitutes for tax-equity financing. Under tax-equity financing, the investor obtains an ownership interest in the project and has incentives to conduct due diligence and monitor its development and operation. Consistent with this, Garrett and Shive (2025) find that wind projects financed by tax-equity investors produce more electricity relative to their generating capacity and attribute this difference to monitoring and contracting. A purchaser of a transferable credit instead acquires the tax benefit without obtaining an ownership interest or operational control. Although credit buyers have incentives to assess credit eligibility and tax risk, they may not provide the same ongoing oversight of project development and operation as an investor with an ownership stake. Thus, if transfers replace tax-equity financing, project operating performance may decline.

Second, transferability may weaken project monitoring to the extent that it substitutes for tax-equity financing. Under tax-equity financing, the investor obtains an ownership interest in the project and has incentives to conduct due diligence and monitor its development and operation. Consistent with this, Garrett and Shive (2025) find that wind projects financed by tax-equity investors produce more electricity relative to their generating capacity and attribute this difference to monitoring and contracting. A purchaser of a transferable credit instead acquires the tax benefit without obtaining an ownership interest or operational control. Although credit buyers have incentives to assess credit eligibility

[18] A related possibility is that the broader transfer market attracts buyers with less experience evaluating clean energy tax credits and the underlying projects. If these buyers conduct less effective screening, lower-quality projects may face fewer barriers to obtaining financing. I examine which firms enter the buyer side of the transfer market in Section 6.1.

and tax risk, they may not provide the same ongoing oversight of project development and operation as an investor with an ownership stake.

H2: *Tax credit transferability has no effect on the quality of clean energy investment.*

4 Research Design and Data Seller-Side

4.1 Research Design

To test H1 and H2, I define treated firms as firms that sell a tax credit and classify them as treated beginning in the year of their first observed transfer. Ideally, I would compare a treated firm's investment after its first transfer with what the same firm would have experienced without the transfer. Because this counterfactual is unobservable, I compare treated firms with firms that operate in the same clean energy sector and are eligible to generate tax credits but have no observed transfer during the sample period.

My baseline analysis estimates the effect of transferring tax credits on investment. Firms first enter the transfer market in different years, and the available follow-up period differs across entry cohorts because my sample ends in 2025. I therefore estimate two specifications that trade off the length of the post-treatment window against the number of treated firms. The first is a standard DiD that compares firms first transferring a credit in 2023 with the comparison firms defined above. For this cohort, I observe three post-treatment years, providing the longest available post-period. Because all treated firms in this specification share a single treatment year, the design avoids comparisons between firms treated at different times.

The second specification extends the analysis to firms first transferring credits in 2024 and 2025 through a stacked cohort design. For each treatment cohort, I construct a separate sample that compares firms first treated in that year with eligible firms that remain untreated during the relevant cohort window. I align each cohort around its first treatment year and combine the cohort-specific samples for estimation. This approach prevents firms that have already transferred a credit from serving as controls for firms treated later and avoids problematic comparisons that can arise in conventional two-way fixed effects models when treatment effects vary across cohorts or over time (Baker et al. 2022; Baker et al. 2026). Stacking doubles the number of treated firms, but the later cohorts contribute fewer post-treatment observations. For both specifications, I estimate the following model:

$$Y_{i,c,t} = \beta_1 \times (Treat_{i,c} \times Post_{c,t}) + \gamma' X_{i,c,t-1} + \lambda_{i,c} + \omega_{t,c} + \varepsilon_{i,c,t} \tag{1}$$

where $Y_{i,c,t}$ is an investment outcome for firm i in cohort stack c and year t. $Treat_{i,c}$ equals one for firms first treated in cohort c, and $Post_{c,t}$ equals one in the treatment year and subsequent years. Cohort-by-firm fixed effects, $\lambda_{i,c}$, absorb time-invariant firm differences within each stack. Cohort-by-year fixed effects, $\omega_{t,c}$, absorb shocks common to firms within each cohort-specific comparison. In the single-cohort specification, the cohort index drops out, and these become firm and year fixed effects. The coefficient of interest, β_1, captures the estimated change in the outcome after a firm's first observed transfer relative to the change among comparison firms.

$\mathbf{X}_{i,c,t-1}$ is a vector of one-year-lagged firm characteristics. For the financial statement analyses, the controls include log total assets, ROA, and leverage. For the electric generator analyses, they include log net generation and the percentage of generation from clean sources. As a robustness test, I replace the lagged controls with pre-treatment averages of the same firm characteristics interacted with year indicators. This alternative specification avoids conditioning on characteristics that may themselves respond to an earlier tax credit transfer while allowing firms with different pre-treatment characteristics to follow different patterns over time.

I measure investment using two outcomes. The first, $CapExInvestment_{i,t}$, is capital expenditures divided by total assets at the end of year $t-1$. The second, $Generators_{i,t}$, is the number of distinct electricity generators operated by firm i in year t and provides a more granular physical measure of electric utilities' generating assets. I classify generators as clean or non-clean based on their underlying technology. Clean generators include, for example, solar, wind, and biomass generators, while non-clean generators include coal- and natural-gas-fired generators. Appendix Table A.2 provides the full classification. I use $\log(Generators_{i,t}+1)$ in the primary specification. Together, the two measures capture complementary financial and physical dimensions of the investment response. As robustness tests, I also examine the logarithm of capital expenditures and estimate a PPML specification for the number of generators (Santos Silva and Tenreyro 2006; Cohn et al. 2022; Chen and Roth 2024).

One concern inherent to the setting is that firms choose whether to transfer their tax credits, raising endogeneity concerns related to differences in treatment and control firms. Firm fixed effects absorb time-invariant differences across firms, while the lagged controls account for differences in observable firm characteristics over time. To further improve comparability between treated and control firms before treatment, I apply entropy balancing separately within each treatment cohort (Hainmueller 2012). I reweight comparison firms so that the pre-treatment means of the balancing variables match those of treated firms. The balancing variables include pre-treatment firm size, ROA, leverage, total generating

capacity, and the clean generation share.[19] The main identifying assumption is that, absent a transfer, outcomes for treated firms would have followed the same trends as outcomes for comparison firms. My event-study plots show no statistically significant differences in pre-treatment investment trends, consistent with the parallel trends assumption.

To test H2, I move from the firm to the project level and examine whether the quality of a firm's generating assets changes after a transfer. I measure quality with the capacity factor, which is realized electricity generation divided by the generation that would occur if the project ran continuously at full capacity, so that a higher value indicates more output per unit of installed capacity. I conduct separate analyses of projects added after the first transfer and projects already in operation. The first test examines whether transferability draws firms toward marginal projects with lower operating performance. The second examines whether proceeds from credit sales improve the operation of existing assets. I use the same 2023-cohort and stacked cohort comparisons, assigning treatment according to the first observed transfer of the firm operating the project. I compare projects operated by transferring firms with projects operated by eligible firms that have no observed transfer during the sample period.

4.2 Identifying Tax Credit Transfers

Because tax credit transfers are new, no existing dataset identifies firm level tax credit transfers. I therefore hand-collect transfers from public disclosures. I first use the SEC EDGAR full-text search to identify public firms that mention tax credit transfers or related terms. For each identified firm, I then review its filings covering 2023 through 2025, including 10-Ks, 10-Qs, 8-Ks, and relevant exhibits, to identify the first year in which it sells or purchases a credit.[20] I supplement these filings with news coverage from Factiva and targeted Google searches. These additional sources allow me to identify transfers involving private firms and to reconcile transaction details across multiple sources. Across the 2023 to 2025 treatment window, I identify 80 firms that sell a transferable credit and 138 firms that purchase one.[21] Where disclosed, I record the credit amount, transaction price, transaction

[19] Appendix Table B.3 compares pre-treatment characteristics across treated firms, the unweighted comparison group, and the entropy-balanced comparison group. Entropy balancing eliminates differences in the matched means. I do not also balance on second moments because of the small number of treated firms. Adding a second-moment constraint for each covariate can concentrate weights on a small number of comparison firms.

[20] I observe a transfer only if at least one party publicly discloses it. Because the amounts involved are material, public firms typically report them, so the sample should capture the economically significant activity in the market. I assess the coverage of my hand-collected sample below.

[21] A single transaction has both a buyer and a seller, which I record separately. I often observe only one party to a transaction and therefore cannot match most buyers and sellers. The larger number of observed buyers

costs, and credit type, although these details are not reported consistently across firms. I use the first observed year in which a firm sells or purchases a transferable credit to define participation and treatment timing.

Because not every transfer is disclosed, I assess how much of the market my hand-collected sample captures. Appendix Figure A.1 Panel (a) compares the aggregate transfer volume by year in my sample against the market estimates reported by Crux, a major tax credit transfer platform that facilitates transfers.[22] My sample captures roughly 70 percent of estimated market volume in 2023, declining to about half by 2025. These figures indicate that my sample captures a substantial share of the dollars transferred in a market for which no complete public record exists.[23] The lower coverage in later years is consistent with the market expanding to private developers and smaller transactions that are less likely to appear in public disclosures. Reporting lags may also affect 2025 because some transfers for the 2025 tax year are first disclosed in filings made during 2026. The median price in my sample is 93 cents per dollar of credit, close to market prices reported by Crux (2025). The average transfer size is $208 million.

Appendix Figure A.1 Panel (b) plots the number of observed sales and purchases by year. Market participation grows rapidly following the first transfers in 2023. I record 17 credit sales in 2023 and 42 in 2025, while the number of purchases increases from 12 in 2023 to 105 in 2025. Investment tax credits are the most frequently transferred credits in the sample, followed by production tax credits.

4.3 Data and Sample Construction

Table 1 reports the construction of the seller-side sample. I begin with the hand-collected sample of firms that disclose selling a transferable tax credit between 2023 and 2025. This sample contains 80 seller firms and 240 seller firm-year observations. I classify these firms as electric utility firms or non-utility firms. My main analysis focuses on electric utility firms because they represent the largest group of treated firms and provide the most comprehensive data for measuring investment outcomes.[24] To study electric utility firms, I merge the tax credit sellers with plant and generation data from the Energy Information

is consistent with many clean energy developers being private and therefore having fewer public disclosure requirements. To the extent that I fail to identify firms that sell or purchase tax credits, these firms may be misclassified as controls. Such misclassifications would make it more difficult to detect differences between treated and comparison firms.

[22] For transfers that do not disclose a transaction price, I impute the price using the mean within credit type and technology group.

[23] The Crux market totals are themselves estimates rather than observed population totals.

[24] I plan to examine non-utility sellers separately. These firms include manufacturers of solar panels and wind tower components as well as clean fuel producers.

Administration. I use Forms EIA-860 and EIA-923 covering 2018 through 2025, with 2018 used only to construct lagged variables, and retain eligible electric utility firms that do not match to an observed seller as the comparison group. I exclude tax-exempt municipal utilities, cooperatives, and public power entities because they do not provide a comparable counterfactual for taxable firms using transferability. I also require sufficient data coverage around the treatment period and non-missing data for the control variables.[25] These restrictions result in a panel of 23,367 firm-year observations for 3,729 public and private electric utility firms, including 45 treated firms and 3,684 control firms.

I construct two analysis samples from this utility panel. The primary sample contains publicly traded electric utility firms and allows me to examine both financial and electricity-based investment outcomes. I restrict the overall utility panel to firms that can be matched to Compustat and have non-missing total assets and sufficient data to construct the control variables. This results in 1,156 firm-year observations for 181 firms over 2019 through 2025, of which 21 are treated. I then construct the cohorts used in the stacked cohort DiD design. Each cohort contains firms first treated in 2023, 2024, or 2025 and control firms that do not transfer a credit during the sample period. I restrict each stack to the event window from four years before through two years after treatment and require at least two pretreatment observations and one observation in or after the treatment year. The resulting public firm sample contains 2,473 firm-cohort-year observations representing 429 firm-cohort pairs. The 2023 cohort sample contains 1,081 firm-year observations for 168 firms, of which 11 are treated.

To broaden the analysis beyond publicly traded firms and increase the number of treated electric utility firms, I construct a second sample that also includes private firms. This increases the number of treated firms from 21 to 45 and brings smaller clean energy firms into the analysis, many of which are privately held and therefore absent from Compustat. The broader sample therefore allows me to examine whether the investment response extends beyond larger publicly traded firms. For this sample, however, I can examine only EIA outcomes because financial statement data are not available for private firms. I apply the same cohort stacking, event window, and coverage requirements as in the public firm sample. The resulting sample contains 58,344 firm-cohort-year observations representing 10,283 firm-cohort pairs. The 2023 cohort sample contains 21,512 firm-year observations for 3,269 firms, of which 19 are treated. This broader sample complements the primary public firm sample by allowing me to examine changes in generating assets

[25] I require at least two observations before treatment and one observation in or after the treatment year. For treated firms, coverage is defined relative to the first transfer year. For control firms, I use 2023 as the reference year.

across a substantially larger and more diverse set of electric utility firms, including smaller privately held firms. I winsorize all continuous non-count variables at the 1st and 99th percentiles.

4.4 Descriptive Statistics

Table 2 reports descriptive statistics for the seller side samples prior to cohort stacking. Panel A reports the primary sample of public electric utility firms. Average capital expenditures are $1.98 billion and equal 6 percent of beginning total assets. The average firm operates roughly 33 generators with a total generating capacity of 3,341 MW. Panel B reports the broader sample of public and private electric utilities. Firms in this sample are substantially smaller, operating an average of approximately 5 generators with 244 MW of capacity. On average, 69 percent of their electricity generation comes from clean energy sources. Appendix Table B.2 reports corresponding descriptive statistics for the seller side samples after cohort stacking, for the full stacked-cohort sample and for the 2023 cohort sample.

5 Results Seller-Side

5.1 Investment Level Responses

Table 3 reports DiD estimates of the capital expenditure response to tax credit transfers across several specifications. Columns (1) and (2) use the stacked cohort design with the unweighted baseline control group. Column (1) includes only cohort-by-firm and cohort-by-year fixed effects, and Column (2) adds one-year-lagged controls. Column (3) restricts the sample to the 2023 cohort, which has the longest available follow-up period, and includes firm and year fixed effects and lagged controls. Columns (4) and (5) repeat the specifications in Columns (2) and (3) using entropy balancing. Across specifications, the coefficient on $Treat \times Post$ is consistently positive, providing evidence of higher investment among transferring firms relative to control firms.

In my preferred specification in Column (4), which uses entropy balancing and lagged controls, capital expenditures increase by 1.2 percentage points of beginning-of-year total assets following a firm's first tax credit transfer. This estimate is statistically significant at the 10 percent level. Relative to the treated firms' pre-treatment mean of 7 percent, it corresponds to an increase of approximately 17 percent. At treated firms' mean pre-treatment lagged assets of $31.6 billion, this translates into approximately $374 million

in additional annual capital spending per firm. The estimate for the 2023 cohort is larger. The entropy-balanced estimate in Column (5) is 2.2 percentage points and statistically significant at the 5 percent level. This corresponds to an increase of approximately 34 percent relative to the treated 2023-cohort firms' pre-treatment mean of 6.5 percent. At their mean pre-treatment assets of $34.7 billion, the estimate implies roughly $770 million in additional annual capital spending per firm. The positive investment response therefore persists, with a larger estimated effect, when I restrict the analysis to the cohort with the longest post-period.

Figure 3 presents the corresponding event-study estimates for the primary sample of public electric utilities in Panel (a) and for the 2023 cohort in Panel (b), using the baseline and entropy-balanced specifications with one-year-lagged controls. In both panels, the pre-treatment coefficients are close to zero, consistent with similar investment trends before the first transfer. Following treatment, the estimates become positive and increase over the observed post-treatment periods. For the 2023 cohort, the entropy-balanced estimates are statistically significant at the 5 percent level in the first two post-treatment years. These patterns are consistent with the positive investment response documented in Table 3. Appendix Table B.4 and Figure B.1 report similar results when replacing the lagged controls with pre-treatment averages interacted with year indicators and when measuring capital expenditures in logs.

I next examine whether the higher capital expenditures are accompanied by an expansion in firms' electric generators, which is the predominant asset purchased by clean energy firms and produces the electricity they sell. Table 4 reports effects on total generators (Panel A), clean generators (Panel B), and non-clean generators (Panel C), with each outcome measured as $\log(\text{count}+1)$. Columns (1) through (4) use the unweighted baseline control group. Columns (1) and (2) use the stacked cohort DiD design for the primary sample of public electric utilities, without and with lagged controls, respectively. Column (3) restricts this sample to the 2023 cohort, while Column (4) extends the analysis to public and private electric utilities. Columns (5) through (7) repeat the specifications in Columns (2) through (4) using entropy balancing.

Panel A shows a positive and statistically significant increase in total generator counts across all specifications. In the preferred public-firm specification, Column (5), the coefficient is 0.119 and is statistically significant at the 1 percent level. Evaluated at the treated firms' pre-treatment mean of roughly 73 generators, this corresponds to an increase of

approximately nine generators.[26] The entropy-balanced estimate for the 2023 cohort in Column (6) is similar at 0.111. The positive response also remains when private electric utilities are included.

Because clean energy tax credits are intended to encourage investment in clean technologies, I next examine which types of generators account for the expansion by separating clean from non-clean generators. The estimated effect on clean generators in Panel B is positive and statistically significant across all specifications. In the preferred entropy-balanced specification with lagged controls, Column (5), the coefficient is 0.138, and 0.148 for the 2023 cohort in Column (6). In contrast, none of the estimated effects on non-clean generators in Panel C is statistically significant. These findings provide evidence of an expansion in clean generators, with no detectable corresponding increase in non-clean generators.

Figure 4 presents the corresponding event-study estimates for the primary public-firm sample, with the entropy-balanced estimates for the 2023 cohort added as a solid line. The pre-treatment coefficients are statistically indistinguishable from zero for all three generator outcomes, consistent with the parallel trends assumption. Following treatment, the estimates increase for total and clean generators, while there is little corresponding movement in non-clean generators. These patterns are consistent with the results in Table 4 and provide additional evidence that the increase in generating assets is concentrated in clean technologies.

Appendix Table B.5 reports the corresponding robustness checks for the generator outcomes. The estimated effects remain similar in sign and magnitude when replacing lagged controls with pre-treatment averages interacted with year indicators and when estimating a PPML specification for generator counts. Together, the financial and physical investment measures provide evidence that firms expand clean energy investment following their first tax credit transfer. These findings support H1, which predicts that tax credit transferability increases the level of clean energy investment.

5.2 Cross-Sectional Tests

I next use cross-sectional tests to examine whether the investment response is consistent with transferability addressing the monetization and financing frictions discussed above. I first examine the monetization channel. If transferability increases investment by allowing firms to realize the value of credits they cannot readily use themselves, the

[26] Because the dependent variable is $\log(G+1)$, I calculate the illustrative change in generator counts as $(e^{\beta}-1)(G+1)$. Evaluating this expression at $\beta = 0.119$ and $G = 72.7$ gives approximately 9.3 additional generators.

response should be more pronounced among firms that pay little or no tax. I proxy for firms' ability to use credits directly using several pretreatment characteristics, including a firm's profitability, the size of its net operating losses, and whether the firm reports a pretax loss. For each proxy, I split the sample into two groups based on the pretreatment characteristic and estimate the specification separately for each subsample. I also report the p-value from a test of whether the estimated treatment effects differ across the two groups.

Panel A of Table 5 reports the results for capital expenditures and clean generators using the entropy-balanced specification with lagged controls. Across these measures, the investment response is generally stronger among firms with a more limited ability to use credits directly. Firms with lower profitability, greater net operating losses, and pretax losses exhibit larger increases in capital expenditures than their counterparts. The clean generator estimates point in the same direction, with the largest difference for firms with pretax losses. I find a statistically statistically significant difference between groups for net operating losses in the capital expenditures response and for pretax losses in the clean generator response. Taken together, these patterns are consistent with the monetization channel, whereby transferability increases investment more when firms have greater difficulty using the credits they generate themselves.[27]

I next examine the financing channel. If transferability increases investment by providing firms with immediate cash and reducing their reliance on external financing, the investment response should be stronger among firms that face greater financing constraints before treatment. I measure financing constraints using pre-treatment leverage, operating cash flow, interest coverage, and the Hadlock–Pierce size-age index (Hadlock and Pierce 2010).

Panel B of Table 5 reports the results. The capital expenditure results are mixed. The response is larger among highly levered firms and firms with low interest coverage, although these differences are not statistically significant. It is not stronger among firms with negative operating cash flow or a high Hadlock–Pierce index. In contrast, the clean generator response is consistently stronger among firms facing greater financing constraints. The between-group differences are statistically significant at the 10 percent level for the operating cash flow and Hadlock–Pierce index splits. These clean generator results are consistent with transferability increasing investment by providing internal funds and reducing reliance on external finance. Together with the monetization tests, the findings provide

[27] Panel A also reports prior access to tax-equity financing as an additional dimension that may affect firms' ability to realize the value of tax incentives. The results are more mixed for this split. Firms with prior tax-equity participation exhibit a larger capital expenditure response, while the clean generator response is similar across the two groups, and neither difference is statistically significant.

suggestive evidence that both channels contribute to the investment response, although the strength of the evidence varies across outcomes and measures.

Finally, I examine whether the investment response varies with firms' existing exposure to clean energy. This analysis examines which types of firms expand investment following a transfer. Firms with less existing clean generation may have greater scope to shift their generating portfolios toward clean energy, while firms already active in clean technologies may be better positioned to deploy additional investment because of their existing expertise and project pipelines. I measure prior clean energy exposure using the firm's pre-treatment share of clean generation and the number of different clean technologies it operates. Appendix Table B.6 reports the results. Overall, the evidence points to stronger investment responses among firms with less prior exposure to clean energy. However, the between-group difference is statistically significant only for the capital expenditures when firms are split by their pre-treatment clean generation share. This finding is relevant to the policy objective because it suggests that transferability can expand clean energy investment beyond firms that were already heavily invested in clean technologies and encourage investment among firms with greater scope to shift their generation toward clean technologies.

5.3 Investment Quality Responses

I next examine whether the increase in investment following a tax credit transfer is accompanied by a change in investment quality. I examine new and existing projects separately. New projects may have lower operating quality if transferability makes marginal projects profitable. Existing projects may perform better if firms use the tax credit sale proceeds to improve their operation. I measure quality at the firm-project level using the capacity factor. I restrict this firm-project sample to solar and wind projects and exclude battery storage, for which capacity factors are not a comparable measures of quality.[28]

Table 6 reports the results for new projects in Panel A and for existing projects in Panel B. In Panel A, I compare the capacity factors of projects commissioned after a firm's first transfer with those commissioned before the transfer, relative to the corresponding change among control firms. For each project, I measure the capacity factor in the first full year after commissioning. In Panel B, I compare changes in the annual capacity factors of projects already operating at the firm's first transfer with the corresponding changes in projects operated by control firms. Columns (1) through (3) use the baseline control group, while Columns (4) and (5) use entropy balancing. Column (1) includes only fixed effects,

[28] For battery storage, the EIA reports net generation as electricity discharged minus electricity used for charging and operation. Because storage involves energy losses, net generation is typically negative. Capacity factors based on this measure are therefore not comparable with those of solar and wind projects.

and Column (2) adds one-year-lagged controls. Column (3) restricts the sample to the 2023 cohort. Columns (4) and (5) repeat the specifications in Columns (2) and (3) using entropy balancing.

For new projects, Panel A shows positive but small and statistically insignificant estimated effects on capacity factors across specifications. If transferability drew in marginal projects with lower operating productivity, I would expect the capacity factors of newly commissioned projects to decline. The estimates provide no evidence of such a decline. Thus, I find no evidence that projects commissioned after the first transfer perform worse than earlier project cohorts, relative to the corresponding change among control firms.

For existing projects, Panel B shows estimated effects that are close to zero and statistically insignificant across specifications, indicating no detectable change in operating performance following a transfer. Viewed alongside the investment-level results, this pattern is consistent with credit-sale proceeds supporting expansion in generating assets without a detectable improvement in the performance of existing projects. Such expansion is consistent with the IRA's goal of increasing clean energy capacity.

Taken together, the results do not provide clear evidence in favor of either of the competing predictions for investment quality. I do not find evidence that the additional resources associated with transferability improve operating performance, but neither do I find evidence that lower-quality marginal projects or weaker monitoring when transfers replace tax-equity financing reduce investment quality. I therefore cannot reject the null hypothesis in H2.

6 Buyer-Side Responses

In the second part of the paper, I descriptively examine what firms participate in and how firms benefit from the new market for transferable tax credits. The buyer-side analysis proceeds in three steps. I first characterize who enters the market, comparing tax credit buyers with pre-IRA tax-equity investors, and examine which firm characteristics predict a purchase. I then quantify the tax benefit of participation by estimating the effect of a tax credit purchase on firms' effective tax rates. Finally, I examine how buyers describe these transactions in their disclosures to explore whether firms use tax credit purchases for broader sustainability objectives in addition to their tax benefits.

6.1 Market Participation

A central policy objective of transferability is to help clean energy firms monetize their tax credits by broadening access to potential buyers. This objective depends not only on developers being able to sell credits but also on other firms' willingness and ability to purchase them. A broader pool of buyers can increase demand for transferable credits, allowing more developers to monetize their credits and obtain funding for clean energy investment. Before the reform, tax-equity partnerships were the primary way for outside investors to finance clean energy projects in exchange for tax benefits. These arrangements required an ownership interest in the underlying project as well as substantial legal, tax, and transaction expertise, concentrating participation among large banks and other financial institutions (Garrett and Shive 2025). Transferability instead allows a firm to purchase a credit directly without acquiring an ownership stake or participating in project operations. These lower entry barriers can broaden and diversify the set of firms willing to finance clean energy. Participation may nevertheless remain concentrated if firms with greater tax expertise, established relationships, and lower due diligence costs retain an advantage in the new market.

I examine market participation in two steps. I first compare firms that purchase transferable tax credits with the tax-equity investors that participated in the market before the IRA. I then compare tax credit buyers with a broader set of potential corporate buyers to examine which characteristics are associated with market entry. For the first analysis, I use my hand-collected sample of 138 firms that publicly disclose purchasing at least one transferable tax credit between 2023 and 2025. I complement these data with tax-equity transactions from the GlobalData Advanced Deals database covering 2015 through 2025 and match both sets of firms by name to Compustat to obtain industry and financial information. Table 7 describes the corresponding sample construction.

I first compare observed activity in the transferable credit market with the tax-equity market that preceded it. Figure 5 Panel (a) plots the number of newly formed tax-equity partnerships and the number of firms that disclose at least one tax credit purchase in each year from 2015 through 2025. Before the IRA, the number of new tax-equity partnerships is relatively stable, ranging from 5 to 12 per year. In the transfer market, 14 firms disclose a purchase in 2023, followed by 66 in 2024 and 126 in 2025.[29] This growth indicates a substantial expansion in disclosed buyer participation in clean energy finance.[30]

[29] Appendix Table C.2 reports the underlying annual counts. Because firms can purchase credits in multiple years, the annual buyer counts sum to 206 firm-year observations but represent 136 unique buyers over the sample period.

[30] Tax-equity partnerships are typically long term, so the number of newly formed partnerships in a given year understates the stock of active tax-equity relationships. I therefore interpret this comparison as evidence on

The buyers entering this market are economically significant. Publicly traded buyers have an average market capitalization of $25.4 billion. Participation also extends across a broader range of industries. Figure 5 Panel (b) reports the industry distribution of tax-equity investors and tax credit buyers. Tax-equity participation is concentrated almost entirely in finance and insurance, whereas tax credit buyers span 13 additional industries. The finance and insurance sector remains the largest among buyers, but manufacturing and retail trade also account for substantial participation. Examples of buyers include Keurig Dr Pepper, Tripadvisor, Best Buy Co., and McDonald's Corporation. The buyer pool therefore includes both firms with a traditional presence in energy finance and firms outside the historical tax-equity investor base. Taken together, the growth in market activity and the expansion across industries are consistent with transferability broadening and diversifying participation in clean energy finance. These comparisons are descriptive and rely on publicly disclosed transactions, so I do not interpret them as causal estimates of the effect of transferability.

Having documented the development of the buyer pool, I examine which firms enter the market. I compare tax credit buyers with corporations that could plausibly have purchased credits but did not disclose a purchase. I begin with Compustat firms operating in the same four-digit SIC industries as observed buyers. This restriction allows me to compare buyers and non-buyers within industries in which at least one firm has entered the market. I require nonmissing financial data and sufficient panel coverage, defined as at least three pre-market observations between 2019 and 2022 and at least two post-market observations between 2023 and 2025. To focus on firms likely to be able to use purchased credits, I further require comparison firms to report positive pretax income. The resulting sample contains 134 firms that purchase a tax credit between 2023 and 2025 and 1,160 comparison firms that do not disclose a purchase.[31]

Table 9 compares the two groups using characteristics measured before the transfer market opened. I organize the characteristics around general firm characteristics, tax payments, tax planning, and environmental orientation. Future buyers are larger than non-buyers, with average log total assets of 9.27 compared with 8.65. They are also older, more leveraged, and more profitable, with an average ROA of 6 percent compared with 4 percent. They also report higher pretax income and cash tax payments. Pretax income averages 8 percent of assets among buyers compared with 6 percent among non-buyers, while cash taxes average 2 percent and 1 percent of assets, respectively. Buyers are also

the scale of new market activity rather than as a direct comparison of the total number of active participants in the two markets. Even with this distinction, the rapid increase in annual tax credit buyer participation points to a substantial expansion relative to the pre-IRA tax-equity market.

[31] Table 8 reports summary statistics for the corresponding firm-year panel, and Appendix Table C.3 for the stacked cohort panel.

more internationally active, with the foreign operations measure averaging 0.53 compared with 0.31. Prior tax-equity participation, book-tax differences, and GAAP and Cash ETRs are similar across the two groups. Among the environmental characteristics, future buyers have a higher average ESG score (49.3 compared with 42.3) and are more likely to have set an emissions reduction target (26 percent compared with 14 percent). These comparisons show that future buyers differ from non-buyers along several dimensions that may affect the benefits and costs of participation.

I next examine which characteristics predict a firm's decision to purchase a tax credit. Table 10 reports the determinants analysis using pre-purchase characteristics. Because firms first enter the market in 2023, 2024, or 2025, I construct separate two-period cohort stacks containing the year immediately before each cohort's purchase year and the purchase year itself. Each stack compares firms first purchasing in that year with firms that have no observed purchase during the sample period. I then combine these cohort-specific samples for estimation. The results show that firm size and profitability are the main predictors of participation. The tax planning measures provide mixed evidence. Firms with a higher federal GAAP ETR are more likely to purchase credits, while larger book-tax differences are associated with a lower probability of purchase. Prior tax-equity participation is not significantly associated with entry. Although future buyers have higher ESG scores and are more likely to have an emissions reduction target in the univariate comparisons, neither measure significantly predicts participation once other firm characteristics are included. Overall, the results suggest that firm size and firms' ability to use credits to reduce their taxes are important considerations in market entry. These findings are consistent with tax savings being an important incentive for entering the transfer market.

6.2 Effective Tax Rates

I next examine the direct tax benefit firms obtain from purchasing transferable credits. A buyer purchases a credit below face value and uses its full face value to reduce federal income taxes. The economic benefit therefore comes from the difference between the credit's face value and its purchase price. In 2024, average prices were approximately 92.5 cents per dollar for investment tax credits and 95 cents for production tax credits (Crux 2025). For example, a firm that pays $0.925 for a $1.00 credit reduces its taxes by $1.00, generating a benefit of $0.075 before transaction costs. This creates a new tax planning opportunity that does not require the buyer to own or operate the underlying clean energy project (Wilde and Wilson 2018). I examine this benefit in two steps. First, I quantify

the purchase benefit using buyers' tax disclosures. Second, I assess whether purchases are associated with a detectable change in firms' overall ETRs.

To quantify the benefit buyers receive, I use information disclosed in their 10-K tax footnotes. For each transaction with sufficient disclosure, I measure the net benefit in one of three ways: the difference between the credit's face value and the price paid, the tax benefit the buyer reports in its tax footnote, or the effect of the purchase on the effective tax rate in the rate reconciliation. I then scale the benefit by pretax income, which expresses it in percentage points of the effective tax rate. Of the disclosed tax credit purchases in my sample, 43 transactions report at least one of the three measures (Appendix Table C.4). Table 11 shows that the average benefit among these purchases is equivalent to a 1.22 percentage point reduction in ETR, with a median of 1.01 percentage points. This benefit is economically meaningful and comparable in magnitude to tax savings associated with international tax planning (Dyreng and Lindsey 2009).

I next examine whether the purchase benefit is reflected in buyers' overall tax outcomes, using both GAAP and Cash ETRs. GAAP ETR is total income tax expense divided by pretax income, while Cash ETR is cash taxes paid divided by pretax income. These measures can reflect the benefit in different periods. GAAP ETR captures the benefit when it is recognized in income tax expense. Cash taxes paid, in contrast, include amounts paid to acquire credits, while reductions in payments to tax authorities or refunds may occur later. Consequently, a purchase can increase measured cash taxes in one year even if it reduces the buyer's total tax-related cash outflows over time. Appendix Table C.5 illustrates this distinction with a numerical example.

To estimate the effect of a purchase on ETRs, I use the buyer and comparison sample described above and construct a stacked cohort panel around firms' first recognized tax credit purchase. I define each buyer's entry year as the fiscal year in which its 10-K first recognizes the purchased credit. Following the seller-side design, I construct separate cohorts for entry years 2023, 2024, and 2025 and compare buyers with firms that have no observed purchase during the sample period. I then combine the cohort-specific samples for estimation. The resulting panel contains 19,523 firm-cohort-years from 119 treated firms and 1,144 comparison firms. I estimate separate effects for the entry year and subsequent years, relative to the pre-treatment period.[32] I report unweighted baseline estimates and estimates that apply entropy balancing separately within each cohort (Hainmueller 2012),

[32]The specification is

$$Y_{i,c,t} = \beta_1(Treat_{i,c} \times Entry_{c,t}) + \beta_2(Treat_{i,c} \times After_{c,t}) + \gamma' X_{i,c,t-1} + \lambda_{i,c} + \omega_{c,s,t} + \varepsilon_{,i,c,t}, \quad (2)$$

where $Y_{i,c,t}$ is GAAP or Cash ETR, $Entry_{c,t}$ equals one in the entry year of cohort c, and $After_{c,t}$ equals one in subsequent years. $\lambda_{i,c}$ and $\omega_{c,s,t}$ denote firm-by-cohort and cohort-by-industry-by-year fixed effects,

reweighting comparison firms so that their pre-entry means of log assets, ROA, leverage, NOL, and pretax income relative to assets match those of the treated firms.[33] Standard errors are clustered by firm.

Table 12 reports the results, with the GAAP ETR in Panel A and the Cash ETR in Panel B. The baseline GAAP ETR estimates in Columns (1) and (2) indicate an entry-year decline of 1.3 percentage points, although neither is statistically significant. With entropy balancing in Column (3), the decline increases to 2.7 percentage points in the year the credit is first recognized. The estimates for subsequent years are small and statistically insignificant in all columns. For Cash ETR, none of the entry-year or after-entry coefficients is statistically significant. The point estimates are positive, which could reflect the timing of credit purchase payments relative to the resulting tax savings. Appendix Figure C.2 presents the corresponding event-study estimates. For both outcomes, the pre-entry coefficients are close to zero and statistically insignificant, consistent with similar trends before entry. The GAAP ETR estimates decline in the entry year and reverts to zero in subsequent years. The Cash ETR estimates show no statistically significant change. Overall, buying a tax credit reduces a firm's GAAP ETR in the year of the purchase, with no detectable effect on Cash ETR. The difference may reflect the timing of the cash payment relative to the credit's use.

6.3 Disclosures

To examine whether buyers obtain benefits beyond direct tax savings, I also study how firms describe their tax credit purchases in sustainability reports. A tax credit purchase can associate a buyer with clean energy investment and provide content that can be incorporated into broader sustainability narratives (Lin et al. 2024). Because the buyer does not own or operate the underlying project, however, the connection between the purchase and the environmental benefits generated by the project, such as emissions reductions, is indirect. This creates scope for firms to present tax credit purchases as evidence of environmental commitment even when the transaction has a limited connection to the firm's own environmental impact, raising concerns related to greenwashing (Heath et al. 2023; Bilicka et al. 2025; Duchin et al. 2025). I therefore collect the sustainability reports published by tax credit buyers for the years in which they purchase a credit and review how firms describe these transactions. Among buyers that publish a sustainability report,

respectively, with industry s defined by four-digit SIC code. $\mathbf{X}_{i,c,t-1}$ contains one-year-lagged log assets, ROA, leverage, and NOL.

[33] Appendix Table C.6 compares the pre-treatment means of these variables across treated firms, the unweighted baseline control group, and the entropy-balanced control group. Entropy balancing eliminates the differences in the matched means.

approximately half describe their tax credit purchases in connection with supporting the energy transition, investing in clean energy, or advancing the firm's broader environmental commitments. In some cases, firms also describe relationships with clean energy firms through which they acquire renewable energy certificates alongside the tax credits.

Table 13 provides examples of these disclosures. MarketAxess, as an example, links a reduction in its federal tax liability to support for companies contributing to the U.S. energy transition. Tractor Supply emphasizes support for U.S. manufacturing and lower-emission transportation fuels, while Tradeweb describes credit purchases as a potential form of indirect infrastructure investment. Stifel explicitly connects its purchases to its own commitment to the energy transition. Together, these examples illustrate how some buyers present tax credit purchases not only as tax transactions but also as environmental investments or evidence of broader environmental commitments.

7 Conclusion

I study how firms respond to the transferability of clean energy tax credits introduced by the Inflation Reduction Act of 2022. Using hand-collected data on credit transfers, I examine both selling and buying firms. I find that sellers increase capital expenditures by 1.2 percentage points of beginning-of-year total assets, or 17 percent of their pre-treatment mean, which amounts to roughly $370 million in additional annual capital spending per firm. Generator counts rise as well, with the increase concentrated in clean technologies. Cross-sectional patterns are consistent with both the monetization and financing channels contributing to the investment response. The additional investment does not come at the cost of quality. I find no change in the operating performance of either newly added or existing generating assets, which suggests that transferability expands investment without pulling firms toward weaker projects.

On the buyer side, I document participation by corporations well beyond the financial institutions that traditionally supplied tax-equity capital. Buyers are larger, more profitable, and pay more in cash taxes than other public firms, and they also have higher ESG scores and are more likely to have set emissions targets. In a determinants test, however, only firm size and high tax payments predict buying a tax credit, consistent with tax savings being an important incentive for participation. Purchasing credits below face value provides a direct economic benefit. For purchases with sufficient disclosure, I estimate an average benefit equivalent to a 1.2-percentage-point reduction in ETR. Consistent with this, buyers' GAAP ETRs decline by 2.7 percentage points in the year the credit is first recognized, with no detectable effect thereafter and no change in Cash ETRs. I also document how buyers

discuss these transactions in their sustainability reports. Among buyers that publish such reports, approximately half present their tax credit purchases as environmental investments or as evidence of broader environmental commitments.

I contribute to research on the real effects of environmental tax policy by examining how firms respond to a change in their ability to monetize tax incentives. Allowing firms to sell credits increases clean energy investment even though tax-equity financing already offered some firms a way to monetize them, which shows that the cost of monetization shapes investment. I also provide some of the first systematic evidence on corporate buyers of clean energy tax credits, documenting their characteristics, market participation, tax outcomes, and sustainability disclosures.

These findings can inform the current policy debate over clean energy tax incentives by showing how firms' ability to realize the value of tax benefits affects investment and participation in clean energy finance. The seller results indicate that transferability increases clean energy investment, while the buyer evidence documents a broader set of corporations participating in clean energy financing through tax credit purchases. More broadly, my paper shows that the effectiveness of a tax incentive depends not only on its value on paper but also on whether firms can monetize it and finance the underlying investment.

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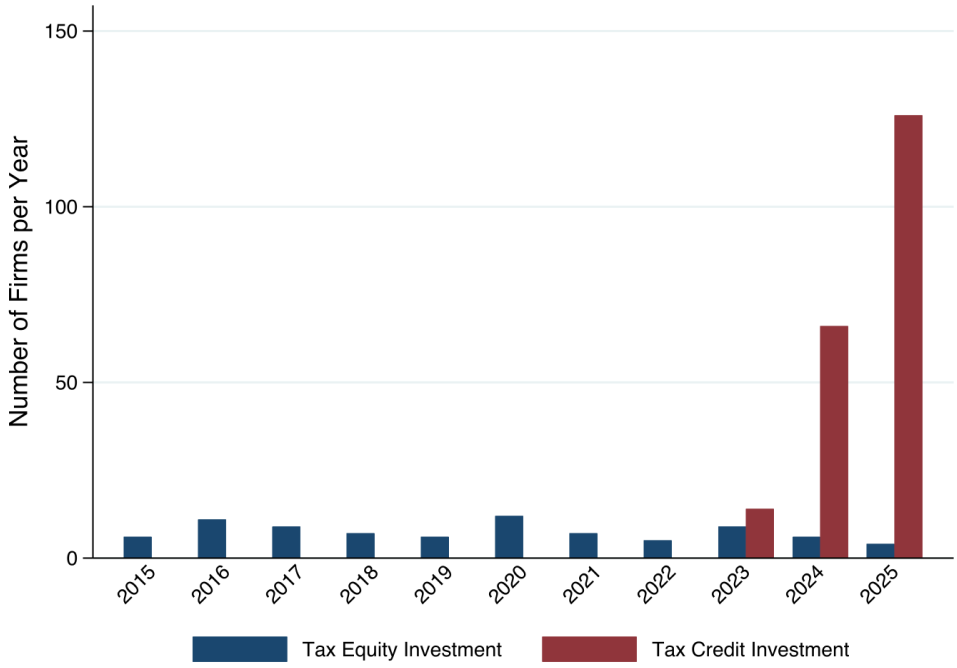
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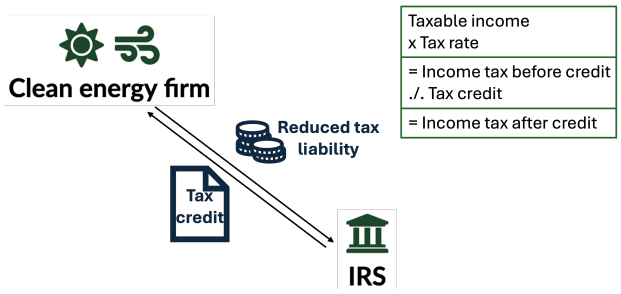
Tables and Figures

Figure 1: Tax Credit Transfer Market

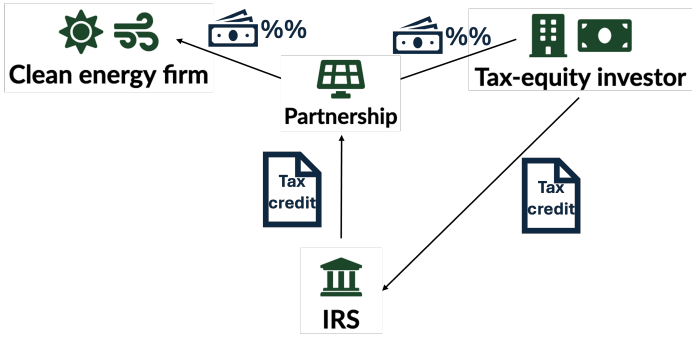


Notes: This figure shows the number of unique firms with tax equity investments (blue) or tax credit investments (red) per year. The sample period is 2015 to 2025. The data is hand-collected from firm disclosures.

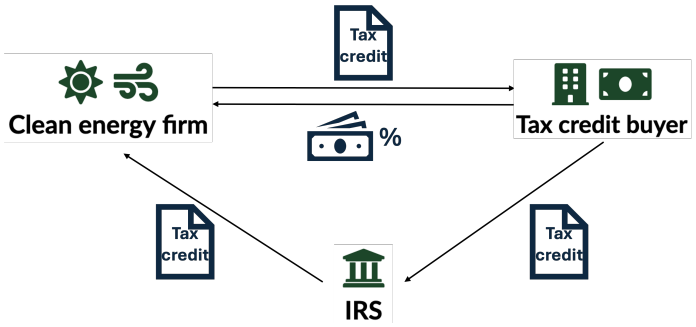
Figure 2: Tax Credit Transferability



(a) Tax credit



(b) Tax equity financing



(c) Tax credit transfer

Notes: This figure illustrates the mechanics of a tax credit transfer under Section 6418 of the Internal Revenue Code, introduced by the Inflation Reduction Act of 2022. Panel (a) depicts a clean energy firm applying a tax credit against its own tax liability. Panel (b) depicts a tax equity financing structure in which a clean energy firm and a tax equity investor form a partnership. Panel (c) depicts a direct transfer, in which a developer lacking sufficient taxable income sells the credit to an unrelated buyer for a discounted cash payment. The buyer claims the full face value against its tax liability, realizing a tax saving equal to the discount. The transaction monetizes the credit without a tax equity partnership and gives the buyer a return without ownership or operational involvement in the underlying project.

Table 1: Seller-Side Responses – Main Sample Construction

	Observations	Firms
Hand-collected tax credit seller sample 2023–2025		
All tax credit seller firm-years	240	80
of which: electric utility firms	150	50
of which: public electric utility firms	98	22
of which: private electric utility firms	52	28
of which: non-utility firms	78	26
Electric utility sample		
Tax credit sellers electric utility firm-years (2023–2025)	150	50
Plus EIA 860/923 electric utility data (2018–2025)	37,757	6,881
Less tax-exempt firms (municipal, co-ops, public power)	(4,403)	(601)
Less firms with insufficient coverage	(6,362)	(2,601)
Less observations with missing data to construct control variables	(1,119)	(1,115)
Public and private electric utility firm-year panel	**23,367**	**3,729**
of which: treated firms	288	45
of which: control firms	23,079	3,684
Sample 1: Public electric utility firms		
Public electric utility parent firm-years with EIA data (2019–2025)	1,366	251
Less observations with missing total assets	(55)	(7)
Less observations with missing data to construct control variables	(13)	(2)
Less firms with insufficient coverage	(142)	(61)
Public electric utility firm-year panel	**1,156**	**181**
of which: treated firms	135	21
of which: control firms	1,021	160
Plus cohort stacking	2,042	320
Less outside event time window [-4, +2]	(458)	(0)
Less insufficient per-stack coverage	(267)	(72)
Public electric utility firm-cohort-year panel	**2,473**	**429**
2023 cohort only	1,081	168
Sample 2: Public and private electric utility firms		
Public and private electric utility firm-year panel	23,367	3,729
Plus cohort stacking	46,158	7,368
Less outside event time window [-4, +2]	(8,148)	(4)
Less insufficient per-stack coverage	(3,033)	(810)
Public and private electric utility firm-cohort-year panel	**58,344**	**10,283**
2023 cohort only	21,512	3,269

Notes: The table reports seller-side sample construction, beginning with the hand-collected tax credit seller sample (2023–2025). For the electric utility samples, I merge U.S. Energy Information Administration data and retain electric utility firms; Sample 1 (public utilities) additionally requires a Compustat match, while Sample 2 (public and private utilities) omits this restriction. Adequate coverage requires at least two pre-years and one post-year relative to the first transfer year (2023 for controls). Each treated cohort forms one cohort stack paired with all never-treated controls over event time $[-4, +2]$.

Table 2: Seller-Side Responses – Summary Statistics for the Firm-Year Panel

	N	Mean	SD	P25	P50	P75
Panel A: *Public electric utility firm-year panel*						
Dependent variables:						
CapEx	1,156	1,979.60	2,922.34	314.34	863.50	2,315.95
CapEx investment	1,156	0.06	0.04	0.03	0.06	0.08
ROA	1,156	0.04	0.08	0.02	0.04	0.06
Generators	1,156	32.94	53.26	3.00	9.00	41.00
Total capacity	1,156	3,340.99	6,706.52	31.90	321.55	3,535.95
Clean generators	1,156	18.16	38.47	0.00	3.00	14.00
Non-clean generators	1,156	14.76	25.40	0.00	3.50	17.00
Capacity factor	1,156	0.39	0.21	0.24	0.38	0.52
Control variables:						
Log(Total assets$_{t-1}$)	1,156	9.72	1.52	8.84	9.65	10.74
ROA$_{t-1}$	1,156	0.05	0.08	0.02	0.04	0.06
Leverage$_{t-1}$	1,156	0.36	0.15	0.29	0.35	0.42
Log(Net generation$_{t-1}$ + 1)	1,156	13.37	3.69	11.32	13.58	16.25
Pct. clean generation$_{t-1}$	1,156	0.39	0.43	0.00	0.15	1.00
Cross-sectional variables:						
NOL/assets	410	0.17	0.38	0.01	0.05	0.15
Pretax income	1,156	2,677.96	7,278.67	147.62	534.72	1,860.00
Ever tax equity	1,156	0.04	0.20	0.00	0.00	0.00
Total assets	1,156	52,025.66	109,816.88	7,213.88	16,474.22	49,886.00
HP index	1,156	-4.33	0.42	-4.64	-4.64	-4.04
OCF	1,155	3,994.15	10,572.26	425.85	1,035.00	3,337.00
Panel B: *Public and private electric utility firm-year panel*						
Dependent variables:						
Generators	23,367	5.30	17.96	1.00	1.00	3.00
Total capacity	23,367	244.01	796.09	3.00	16.00	107.90
Clean generators	23,367	3.39	14.30	1.00	1.00	2.00
Non-clean generators	23,367	1.91	8.50	0.00	0.00	1.00
Capacity factor	23,367	0.31	0.22	0.17	0.24	0.43
Control variables:						
Net generation$_{t-1}$	23,367	786,514.65	2,754,134.95	4,590.00	30,836.00	281,167.00
Pct. clean generation$_{t-1}$	23,367	0.69	0.46	0.00	1.00	1.00

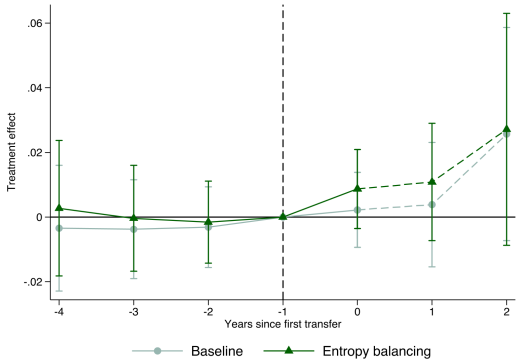
Notes: The table reports summary statistics on the raw firm-year panel, before cohort stacking. Panel A shows summary statistics for Sample 1 (public electric utility firm-year panel) and Panel B for Sample 2 (public and private electric utility firm-year panel). Financial variables are in $ millions, generation is in GWh and capacity in MW. All variables are defined in Appendix B.1.

Table 3: Seller-Side Responses – Effect of Tax Credit Transfer on Investment Level

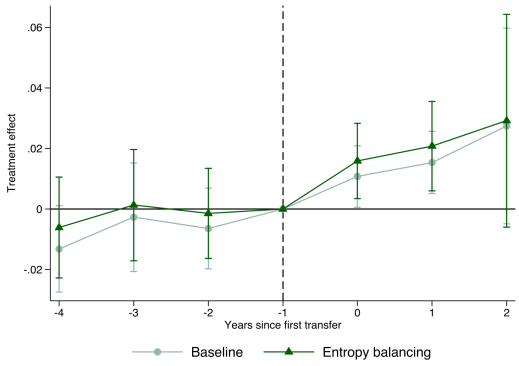
	(1)	(2)	(3)	(4)	(5)
	Baseline			Entropy balanced	
	CapEx investment				
Treat × Post	0.007	0.008	0.021***	0.012*	0.022**
	(1.01)	(1.04)	(2.63)	(1.67)	(2.49)
Log(Total assets)$_{t-1}$		-0.012*	-0.016**	0.009	-0.011
		(-1.66)	(-2.22)	(0.64)	(-0.78)
Pretax income / assets$_{t-1}$		0.004	-0.008	0.032	-0.028
		(0.21)	(-0.38)	(1.24)	(-0.85)
Leverage$_{t-1}$		-0.014	-0.009	-0.009	-0.043**
		(-0.94)	(-0.57)	(-0.48)	(-2.28)
Observations	2,473	2,473	1,081	2,473	1,081
Adj. R^2	0.745	0.747	0.739	0.619	0.670
Sample	Public	Public	2023	Public	2023
Cohort × firm FE	X	X		X	
Cohort × year FE	X	X		X	
Firm FE			X		X
Year FE			X		X
Lagged controls		X	X	X	X

Notes: This table reports stacked-cohort difference-in-differences estimates of the effect of tax credit transfers on investment. The dependent variable is CapEx scaled by one-year-lagged total assets. Columns (1) through (3) use the baseline control group, whereas columns (4) and (5) use entropy balancing. Within the baseline set, the first column includes only cohort × firm and cohort × year fixed effects and uses Sample 1, the public electric utility firm-cohort-year panel; the second adds one-year-lagged control variables, also on Sample 1; the third restricts Sample 1 to the 2023 cohort, the only cohort with a full post-treatment window, with lagged controls and firm and year fixed effects in place of the cohort-interacted terms. The entropy-balanced set repeats the lagged-control specification (column (4)) and the 2023-cohort specification (column (5)); the entropy-balanced, no-controls specification is omitted. All variables are defined in Appendix B.1. t-statistics based on standard errors clustered at the firm level are in parentheses. *** p<0.01, ** p <0.05, * p<0.1.

Figure 3: Seller-Side Responses – Effect of Tax Credit Transfer on Investment Level, Event Studies



(a) CapEx investment



(b) CapEx investment, 2023 cohort

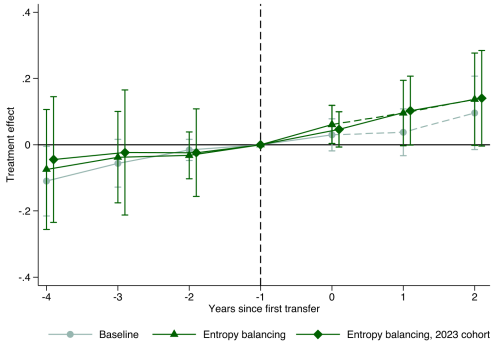
Notes: This figure plots event-study estimates of the effect of tax credit transfers on investment. The estimates are obtained by replacing the *Post* indicator with separate relative-year indicators. The dependent variable is CapEx scaled by one-year-lagged total assets. Panel (a) uses Sample 1, the public electric utility firm-cohort-year panel, with cohort $\times$ firm and cohort $\times$ year fixed effects; Panel (b) restricts Sample 1 to the 2023 cohort, the only cohort with a full post-treatment window, with firm and year fixed effects in place of the cohort-interacted terms. Each regression includes one-year-lagged control variables. Each panel presents estimates using the baseline control group (turquoise circles) and entropy balancing (dark green triangles). In Panel (a), the series are dashed at event years $t+1$ and $t+2$ because the 2024 and 2025 cohorts are only partially represented or not represented at these horizons. Capped vertical lines indicate 95% confidence intervals. Standard errors are clustered at the firm level.

Table 4: Seller-Side Responses – Effect of Tax Credit Transfer on Generator Investment

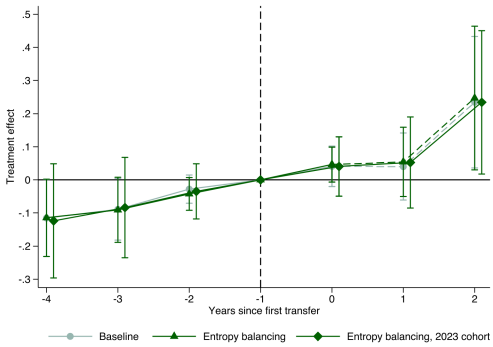
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Baseline				Entropy balanced		
Panel A:	*Log(Generators + 1)*						
Treat × Post	0.083**	0.084**	0.098**	0.066**	0.119***	0.111*	0.048*
	(2.46)	(2.49)	(2.20)	(2.51)	(2.66)	(1.88)	(1.71)
Log(Net gen + 1)$_{t-1}$		0.022**	0.031***	0.014***	0.049	0.104***	0.068***
		(2.38)	(2.65)	(7.23)	(1.48)	(3.37)	(3.87)
Pct clean gen$_{t-1}$		0.171	0.113	-0.061***	0.127	0.300	-0.292*
		(1.07)	(0.62)	(-3.65)	(0.41)	(0.75)	(-1.94)
Observations	2,473	2,473	1,081	58,344	2,473	1,081	58,344
Adj. R^2	0.984	0.984	0.982	0.983	0.978	0.971	0.990
Panel B:	*Log(Clean generators + 1)*						
Treat × Post	0.124**	0.118**	0.166**	0.128***	0.138**	0.148*	0.078*
	(2.14)	(2.11)	(2.06)	(3.54)	(2.30)	(1.66)	(1.95)
Log(Net gen + 1)$_{t-1}$		-0.001	0.003	0.008***	-0.009	0.035*	0.043**
		(-0.07)	(0.15)	(3.87)	(-0.43)	(1.72)	(2.21)
Pct clean gen$_{t-1}$		0.489**	0.447**	0.014	0.652***	1.185***	0.002
		(2.57)	(2.29)	(0.52)	(2.97)	(5.06)	(0.01)
Observations	2,473	2,473	1,081	58,344	2,473	1,081	58,344
Adj. R^2	0.983	0.984	0.982	0.980	0.987	0.983	0.987
Panel C:	*Log(Non-clean generators + 1)*						
Treat × Post	-0.012	-0.007	0.031	0.000	0.021	0.059	0.005
	(-0.26)	(-0.14)	(0.69)	(0.02)	(0.34)	(0.84)	(0.45)
Log(Net gen + 1)$_{t-1}$		0.029***	0.038***	0.007***	0.034	0.075	0.010***
		(2.91)	(2.99)	(4.30)	(0.75)	(1.65)	(2.68)
Pct clean gen$_{t-1}$		-0.108	-0.175	-0.083***	-0.041	-0.394	-0.119**
		(-0.56)	(-0.81)	(-3.43)	(-0.08)	(-0.67)	(-2.04)
Observations	2,473	2,473	1,081	58,344	2,473	1,081	58,344
Adj. R^2	0.981	0.981	0.979	0.992	0.976	0.969	0.998
Sample	Public	Public	2023	+ Private	Public	2023	+ Private
Cohort × firm FE	X	X		X	X		X
Cohort × year FE	X	X		X	X		X
Firm FE			X			X	
Year FE			X			X	
Lagged controls		X	X	X	X	X	X

Notes: This table reports stacked-cohort difference-in-differences estimates of the effect of tax credit transfers on the number of generators, all as log(count + 1): overall generators (Panel A), and split into clean (Panel B) and non-clean (Panel C) generators. Columns (1) through (4) use the baseline control group, whereas columns (5) through (7) use entropy balancing. Within the baseline set, the first column includes only cohort × firm and cohort × year fixed effects and uses Sample 1, the public electric utility firm-cohort-year panel; the second adds one-year-lagged control variables, also on Sample 1; the third restricts Sample 1 to the 2023 cohort, the only cohort with a full post-treatment window, with lagged controls and firm and year fixed effects in place of the cohort-interacted terms; the fourth uses the same lagged-control specification as the second column but is estimated on Sample 2, the public and private electric utility firm-cohort-year panel. The entropy-balanced set repeats the lagged-control, 2023-cohort, and Sample 2 specifications (columns (5)-(7)); the entropy-balanced, no-controls specification is omitted. All variables are defined in Appendix B.1. t-statistics based on standard errors clustered at the utility level are in parentheses. *** p<0.01, ** p <0.05, * p<0.1.

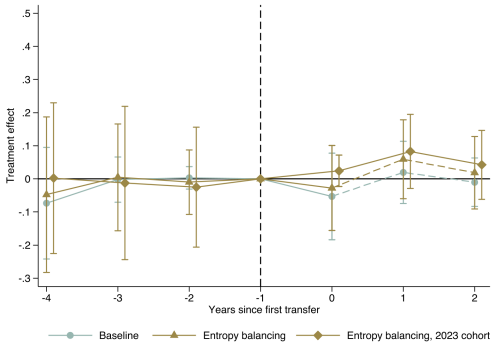
Figure 4: Seller-Side Responses – Effect of Tax Credit Transfer on Generator Investment, Event Studies



(a) Log(Generators + 1)



(b) Log(Clean generators + 1)



(c) Log(Non-clean generators + 1)

Notes: This figure plots event-study estimates of the effect of tax credit transfers on the number of generators: overall (Panel (a)), and split into clean (Panel (b)) and non-clean (Panel (c)) generators, all as log(count + 1), obtained by replacing the *Post* indicator with separate relative-year indicators. All panels use Sample 1, the public electric utility firm-cohort-year panel, and include cohort × firm and cohort × year fixed effects and one-year-lagged control variables. Each panel presents the baseline control group (turquoise circles), entropy balancing (triangles), and entropy balancing restricted to the 2023 cohort (diamonds, solid line), the only cohort with a full post-treatment window; the latter uses firm and year fixed effects in place of the cohort-interacted terms and is shifted slightly to the right of each event year for visibility. The entropy-balanced series are shown in dark green for Panels (a)-(b) and dark brown for Panel (c). The pooled series are dashed at event years $t+1$ and $t+2$ because the 2024 and 2025 cohorts are only partially represented or not represented at these horizons. Capped vertical lines indicate 95% confidence intervals; standard errors are clustered at the firm level.

Table 5: Seller-Side Responses – Cross-Sectional Tests of Investment Response Channels

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Entropy balanced							
Panel A:	*Monetization-Constraint Channel*							
	ROA		NOL/assets		Pretax income		Ever tax equity	
	Low	High	High	Low	Loss	Profit	No	Yes
CapEx investment								
Treat × **Post**	0.016*	0.004	0.070***	-0.012	0.030*	0.006	0.009	0.041***
	(1.83)	(0.35)	(5.49)	(-1.09)	(1.81)	(0.88)	(1.34)	(10.13)
p-value, difference	0.680		<0.001***		0.507		0.258	
Observations	815	1,658	310	623	212	2,261	2,384	89
Adj. R^2	0.605	0.613	0.911	0.869	0.653	0.617	0.636	0.953
Log(Clean generators + 1)								
Treat × **Post**	0.165**	0.081	0.192***	0.154***	0.444***	0.054	0.149**	0.169***
	(2.35)	(0.85)	(3.85)	(3.37)	(2.86)	(0.91)	(2.32)	(4.31)
p-value, difference	0.579		0.931		0.037**		0.162	
Observations	815	1,658	310	623	212	2,261	2,384	89
Adj. R^2	0.992	0.978	0.999	0.998	0.995	0.981	0.986	0.991
Panel B:	*Financing-Friction Channel*							
	Leverage		Oper. CF		Int. coverage		HP index	
	High	Low	Neg.	Pos.	Low	High	High	Low
CapEx investment								
Treat × **Post**	0.022**	-0.000	0.006	0.014**	0.019*	0.004	0.009	0.015
	(2.51)	(-0.01)	(0.59)	(2.01)	(1.91)	(0.56)	(0.99)	(1.61)
p-value, difference	0.175		0.064*		0.249		0.222	
Observations	820	1,653	107	2,358	802	1,628	823	1,650
Adj. R^2	0.721	0.627	0.855	0.586	0.658	0.583	0.765	0.472
Log(Clean generators + 1)								
Treat × **Post**	0.171*	0.087	0.236***	0.110	0.195***	0.015	0.266***	0.047
	(1.97)	(1.36)	(28.84)	(1.60)	(2.65)	(0.22)	(2.81)	(0.70)
p-value, difference	0.262		0.065*		0.227		0.057*	
Observations	820	1,653	107	2,358	802	1,628	823	1,650
Adj. R^2	0.992	0.982	0.998	0.983	0.992	0.980	0.990	0.985
Sample	Public	Public	Public	Public	Public	Public	Public	Public
Cohort × firm FE	X	X	X	X	X	X	X	X
Cohort × year FE	X	X	X	X	X	X	X	X
Lagged controls	X	X	X	X	X	X	X	X

Notes: This table reports the Treat × Post coefficient from separate subsample regressions using Sample 1, the public electric utility firm-cohort-year panel. The dependent variables are CapEx scaled by one-year-lagged total assets and log(Clean generators + 1). All columns use entropy balancing and one-year-lagged controls, omitting any control that mechanically overlaps with the column's split variable. Panel A splits firms by proxies for the monetization constraint and Panel B by proxies for the financing friction. Int. coverage is EBITDA over interest expense, and the HP index is the Hadlock and Pierce (2010) size-age index. ROA, NOL/assets, Pretax income, and Leverage are each firm's pre-treatment mean; Oper. CF, Int. coverage, and HP index are measured in event year $t-1$. Tercile cutoffs are computed on the baseline sample. The row "p-value, difference" reports the p-value on the Treat × Post × group interaction from a pooled regression on the two subsamples. All variables are defined in Appendix B.1. t-statistics based on standard errors clustered at the firm level are in parentheses. *** p<0.01, ** p <0.05, * p<0.1.

Table 6: Seller-Side Responses – Effect of Tax Credit Transfer on Investment Quality

	(1)	(2)	(3)	(4)	(5)
	Baseline			Entropy balanced	
Panel A:	*Capacity factor of new projects*				
Treat × **Post**	0.028	0.031	0.029	0.018	0.016
	(1.02)	(1.08)	(1.06)	(0.75)	(0.68)
Log(Net gen + 1)$_{t-1}$		0.006***	0.006***	-0.016	0.001
		(3.41)	(3.97)	(-1.01)	(0.20)
Pct clean gen$_{t-1}$		0.099	0.005	-0.115	-0.234
		(0.77)	(0.03)	(-0.69)	(-1.53)
Observations	794	794	393	794	393
Adj. R^2	0.733	0.735	0.733	0.856	0.872
Panel B:	*Capacity factor of existing projects*				
Treat × **Post**	0.000	0.000	0.002	0.007	0.015
	(0.03)	(0.03)	(0.21)	(0.82)	(1.40)
Log(Net gen + 1)$_{t-1}$		-0.000	-0.002	0.015***	0.015**
		(-0.01)	(-1.14)	(2.69)	(2.18)
Pct clean gen$_{t-1}$		0.020	0.056	0.078	0.072
		(0.81)	(1.61)	(0.94)	(0.71)
Observations	13,717	13,579	6,025	13,579	6,025
Adj. R^2	0.810	0.813	0.803	0.846	0.834
Sample	Public	Public	2023	Public	2023
Cohort × firm or project FE	X	X	X	X	X
Cohort × technology × year FE	X	X	X	X	X
Lagged controls		X	X	X	X

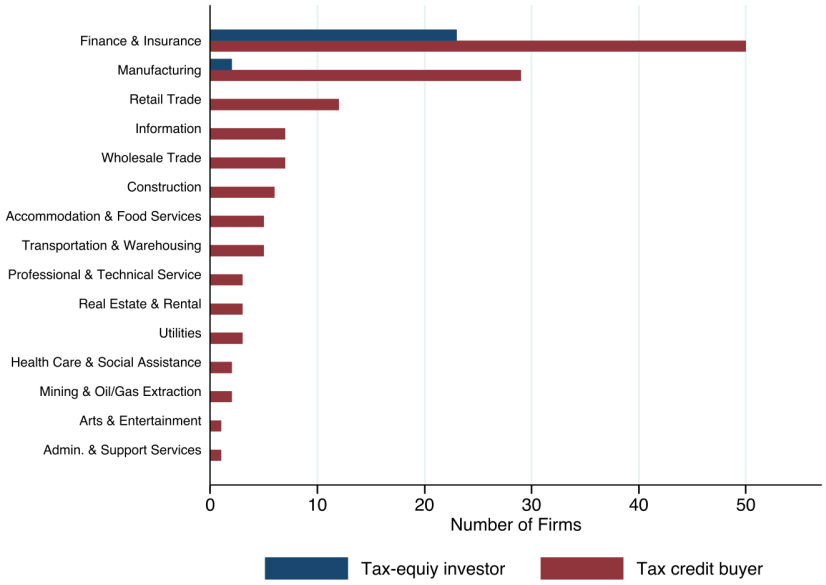
Notes: This table reports stacked cohort difference-in-differences estimates of the effect of tax credit transfers on project capacity factors. The sample is the public electric utility panel (Sample 1), disaggregated to the project level and restricted to solar and wind projects. Panel A reports estimates for new projects, comparing the capacity factor in the first full year of operation for projects commissioned in or after the first transfer year with that of projects commissioned earlier. The Panel A sample is restricted to utilities with at least one project commissioned before the first transfer. Panel B reports estimates for existing projects, comparing annual capacity factors before and after the first transfer for projects already operating at least five years beforehand. All specifications include [fixed effects]. Columns (1) through (3) use the baseline control group. Column (1) includes fixed effects only, column (2) adds one-year-lagged controls, and column (3) restricts the sample to the 2023 cohort. Columns (4) and (5) repeat the specifications in columns (2) and (3) with entropy balancing on the utility's pre-treatment capacity, number of generators, and wind share of pre-transfer projects. All variables are defined in Appendix B.1. t-statistics based on standard errors clustered at the utility level are in parentheses. *** p<0.01, ** p <0.05, * p<0.1.

Table 7: Buyer-Side Responses – Sample Selection

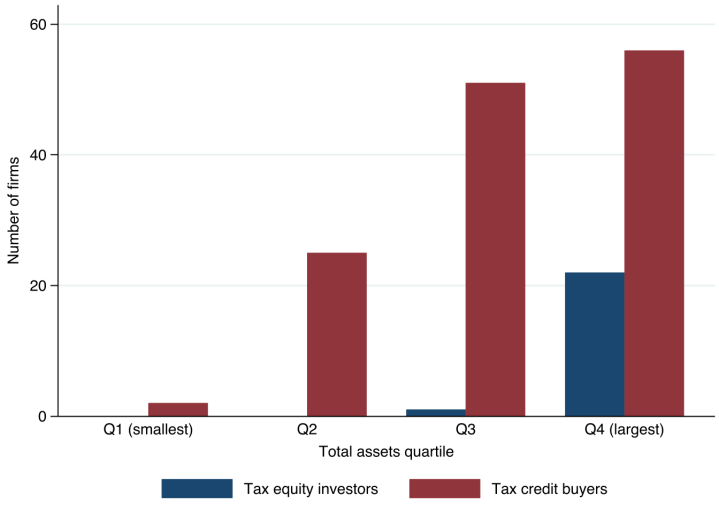
	Observations	Firms
Hand-collected tax credit buyer sample 2023–2025		
All tax credit buyer firm-years	208	138
Sample 1: Tax credit buyer and tax equity firm sample		
Tax credit buyer firm-years (2023–2025)	414	138
Plus tax equity investor indicator (Global Data Advanced Deals, 2015–2025)	396	36
Plus Compustat data (2018–2025)	1,380	174
Less no Compustat SIC match	(112)	(14)
Tax credit buyer and tax equity firm sample	**1,268**	**160**
of which: tax credit buyer firms	1,076	136
of which: tax equity investor firms	216	27
including 3 firms that are both tax credit buyers and tax-equity investors		
Sample 2: Tax credit buyer firm-year panel		
Tax credit buyer and tax equity firm sample	**1,268**	**160**
Plus control firm-years (same 4-digit-SIC controls, 2018–2025)	16,365	2,377
Less firm-years without Compustat financials	(618)	(74)
Less firms with missing or non-positive pretax income	(6,361)	(1,084)
Less year 2018 (used for lagged control variables only)	(1,296)	(0)
Less insufficient coverage	(246)	(61)
Less observations with missing data to construct variables	(394)	(2)
Tax credit buyer firm-year panel (Sample 2, Pre-period comparison)	**8,542**	**1,294**
of which: treated firms	890	134
of which: control firms	7,652	1,160
Sample 3 and 4: Tax credit buyer stacked firm-cohort-year panel		
Tax credit buyer firm-year panel	**8,542**	**1,294**
Plus cohort stacking	15,304	2,320
Less outside event time window [-4, +2]	(3,370)	(2)
Less insufficient per-stack coverage	(953)	(248)
Tax credit buyer stacked panel (Sample 4, ETR analysis)	**19,523**	**3,364**
Tax credit buyer stacked two-period panel (Sample 3, Determinants)	**6,674**	**3,337**

Notes: This table reports the sample construction for the hand-collected tax credit buyer sample. Sample 1 (market participation figures) begins with 138 firms disclosing a tax credit purchase between 2023 and 2025, adds tax equity investors observed between 2015 and 2025, and requires a Compustat SIC code match. Sample 2 (cross-sectional determinants and participation tests) adds control firms in the same four-digit SIC industries as observed buyers and requires nonmissing Compustat financials and at least three years between 2019 and 2022 and two years between 2023 and 2025; control firms must also report positive pretax income in at least three of the four years between 2019 and 2022, and all observations require nonmissing one-year-lagged log total assets, ROA, leverage, and NOL as well as nonmissing GAAP and Cash ETR. Samples 3 and 4 stack each treatment cohort (2023, 2024, or 2025) over an event window of four years before through two years after purchase: Sample 3 (determinants analysis) retains only the purchase year and the year before it, while Sample 4 (effective tax rate difference-in-differences analysis) retains the full firm-cohort-year panel and defines each cohort by the year in which the buyer's 10-K first recognizes the purchased credit.

Figure 5: Buyer-Side Responses



(a) Industry distribution of tax credit and tax equity investment



(b) Firm size

Notes: Panel (a) shows the number of unique firms with tax equity investments (blue) or tax credit purchases (red) per year. Panel (b) shows the number of unique firms per NAICS 2-digit sector for tax equity investors (blue) and tax credit purchasers (red), sorted by total firms (descending); Manufacturing groups NAICS 31–33, Retail Trade groups 44–45, and Transportation & Warehousing groups 48–49. The sample period is 2015 to 2025. The data is hand-collected from firm disclosures.

Table 8: Buyer-Side Responses – Summary Statistics for the Firm-Year Panel

	N	Mean	SD	P25	P50	P75
Firm-year panel (Sample 2)						
Dependent variables:						
Tax credit purchase	8,542	0.02	0.15	0.00	0.00	0.00
GAAP ETR	8,542	0.21	0.14	0.15	0.21	0.24
Cash ETR	8,542	0.20	0.17	0.09	0.18	0.25
Control variables:						
Log(Total assets$_{t-1}$)	8,542	8.74	2.11	7.39	8.61	10.02
ROA$_{t-1}$	8,542	0.04	0.08	0.01	0.02	0.06
Leverage$_{t-1}$	8,542	0.19	0.21	0.02	0.11	0.31
NOL$_{t-1}$	8,542	0.04	0.20	0.00	0.00	0.01
Cross-sectional variables:						
Log(Total assets)	8,542	8.83	2.09	7.48	8.70	10.10
Log(Market cap)	8,080	7.88	2.20	6.25	7.92	9.47
Firm age	8,542	26.60	18.15	12.00	23.00	35.00
Cash / assets	8,542	0.11	0.13	0.03	0.07	0.14
ROA	8,542	0.04	0.08	0.01	0.02	0.06
Leverage	8,540	0.20	0.21	0.02	0.12	0.31
Pretax income / assets	8,542	0.05	0.09	0.01	0.03	0.08
Cash taxes / assets	8,542	0.01	0.02	0.00	0.00	0.02
Persistent profitability	8,542	3.78	0.43	4.00	4.00	4.00
NOL	8,542	0.04	0.18	0.00	0.00	0.01
Tax equity, ever	8,542	0.02	0.14	0.00	0.00	0.00
Book-tax difference	8,542	-0.01	0.48	-0.01	-0.00	0.00
Foreign operations	8,542	0.34	0.47	0.00	0.00	1.00
ESG score	5,592	45.22	19.04	30.39	43.22	60.20
Emission red. target	6,288	0.17	0.38	0.00	0.00	0.00

Notes: The table reports summary statistics on the firm-year panel (Sample 2), before cohort stacking. Control variables are measured in year $t-1$. All variables are defined in Appendix C.1.

Table 9: Buyer-Side Responses – Treated vs. Control: Pre-period Characteristics

	Treated				Control				
	N	Mean	SD	Median	N	Mean	SD	Median	Diff.
Panel A:	*Firm characteristics*								
Log(Total assets)	132	9.27	1.62	9.18	1,147	8.65	2.16	8.55	0.62***
Log(Market cap)	131	8.83	1.84	8.88	1,086	7.71	2.15	7.72	1.12***
Firm age	132	30.60	21.52	27.50	1,147	24.19	17.58	21.50	6.41***
Cash / assets	132	0.10	0.12	0.06	1,147	0.12	0.13	0.08	-0.02
Leverage	132	0.27	0.23	0.24	1,147	0.18	0.20	0.11	0.09***
ROA	132	0.06	0.05	0.04	1,147	0.04	0.06	0.02	0.02***
Panel B:	*Tax liability*								
Pretax income / assets	132	0.08	0.07	0.06	1,147	0.06	0.07	0.03	0.02***
Cash taxes / assets	132	0.02	0.02	0.01	1,147	0.01	0.01	0.01	0.01***
Persistent profitability	132	3.56	0.84	4.00	1,147	3.63	0.68	4.00	-0.07
NOL	132	0.02	0.08	0.00	1,147	0.04	0.16	0.00	-0.02
Panel C:	*Tax planning*								
Tax equity	132	0.02	0.15	0.00	1,147	0.01	0.11	0.00	0.01
GAAP ETR	132	0.20	0.08	0.21	1,147	0.20	0.10	0.20	-0.00
Cash ETR	132	0.19	0.11	0.19	1,147	0.19	0.12	0.19	0.00
Book-tax difference	132	-0.00	0.03	-0.00	1,147	-0.00	0.07	-0.00	0.00
Foreign operations	132	0.53	0.48	0.75	1,147	0.31	0.45	0.00	0.22***
Uncertain tax positions	–	–	–	–	–	–	–	–	–
Panel D:	*Environmental orientation*								
ESG score	120	49.31	19.47	50.04	711	42.31	18.13	39.69	7.00***
Emission red. target	125	0.26	0.40	0.00	812	0.14	0.33	0.00	0.12***

Notes: The table reports pre period (2019–2022) firm characteristics for treated and control firms in the estimation sample, grouped into firm characteristics, tax liability, tax planning, and environmental orientation. Diff. is the treated-minus-control difference from a two-sample t-test. *** p<0.01, ** p<0.05, * p<0.1.

Table 10: Buyer-Side Responses – Determinants of Tax Credit Purchase

	(1) Firm char.	(2) Tax liability	(3) Tax planning	(4) Environmental	(5) Liability + Env.	(6) Planning + Env.
	Tax credit purchase					
Log(Total assets$_{t-1}$)	0.092***	0.092***	0.172***	0.092***	0.099***	0.178***
	(4.60)	(4.50)	(5.19)	(2.71)	(2.75)	(4.09)
ROA$_{t-1}$	3.781***	2.090**	4.355***	3.635***	2.034*	4.091***
	(5.21)	(2.30)	(4.77)	(4.03)	(1.78)	(4.05)
Leverage$_{t-1}$	0.352*	0.307	0.091	0.163	0.144	-0.109
	(1.79)	(1.56)	(0.36)	(0.72)	(0.65)	(-0.40)
Cash / assets$_{t-1}$	-0.695	-0.807	-0.712	-0.766	-0.871	-1.215*
	(-1.31)	(-1.44)	(-1.15)	(-1.26)	(-1.42)	(-1.72)
NOL$_{t-1}$		-0.544			-0.461	
		(-0.88)			(-0.70)	
Cash taxes / assets$_{t-1}$		7.717**			7.235*	
		(2.42)			(1.82)	
Federal GAAP ETR$_{t-1}$			0.951***			0.714**
			(3.05)			(1.99)
Book-tax difference$_{t-1}$			-1.433***			-1.406***
			(-4.00)			(-3.82)
Tax equity			-0.482			-0.598
			(-0.91)			(-1.09)
ESG score$_{t-1}$				0.001	0.000	0.000
				(0.26)	(0.13)	(0.09)
Emission red. target$_{t-1}$				0.217	0.204	0.131
				(1.63)	(1.51)	(0.87)
Observations	3,337	3,337	1,656	2,203	2,203	1,241
Pseudo R^2	0.083	0.088	0.103	0.089	0.093	0.111
Cohort FE	X	X	X	X	X	X

Notes: The table reports probit estimates of the probability of a tax credit purchase, using a stacked cohort panel restricted to the year before and the year of each cohort's purchase, with control firms replicated once per cohort stack. Column (1) includes baseline firm characteristics (log assets, ROA, leverage, cash/assets); Columns (2)–(4) add tax liability, tax planning, and environmental variables in turn; Columns (5) and (6) report combinations of these groups. Fixed effects are cohort fixed effects. The year before each purchase year is perfectly predicted and dropped, so estimation uses one purchase-year observation per firm-cohort pair. t-statistics clustered by firm in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table 11: Buyer-Side Responses – Tax Benefit of Tax Credit Purchase

	Observations	Mean	SD	P25	P50	P75
ETR benefit calculation (pp of pretax income)						
Face value less price paid	18	0.97	0.67	0.49	0.78	1.22
Tax benefit reported in the tax note	37	1.24	1.15	0.55	1.01	1.45
Rate-reconciliation effect	22	1.28	0.59	0.80	1.25	1.70
Combined tax benefit measure	43	1.22	1.09	0.55	1.01	1.45

Notes: The table reports the tax benefit in percentage points of pretax income, which equals percentage points of the effective tax rate. The combined benefit measure uses the reported tax benefit where available, then face value less price, then the rate-reconciliation effect. The sample contains transactions with positive pretax income and at least one benefit measure.

Table 12: Buyer-Side Responses – Effect of Tax Credit Purchase on ETR

	Baseline		Entropy balanced
	(1)	(2)	(3)
Panel A:	*GAAP ETR*		
Treat × Entry year	-0.013	-0.013	-0.027***
	(-1.24)	(-1.30)	(-2.71)
Treat × After entry	0.010	0.008	-0.015
	(0.59)	(0.44)	(-1.23)
Log(Total assets$_{t-1}$)		0.020**	0.022**
		(2.21)	(2.16)
ROA$_{t-1}$		0.067	0.152***
		(0.73)	(3.01)
Leverage$_{t-1}$		0.083**	0.025
		(2.18)	(0.77)
NOL$_{t-1}$		-0.041*	0.016
		(-1.69)	(0.84)
Observations	19,523	19,523	19,523
Adj. R^2	0.198	0.201	0.433
Panel B:	*Cash ETR*		
Treat × Entry year	0.013	0.009	0.016
	(1.01)	(0.72)	(1.43)
Treat × After entry	0.017	0.009	0.019
	(0.81)	(0.42)	(1.09)
Log(Total assets$_{t-1}$)		0.039***	0.040***
		(3.77)	(3.91)
ROA$_{t-1}$		0.125**	0.243***
		(2.31)	(2.83)
Leverage$_{t-1}$		-0.073*	-0.067
		(-1.69)	(-1.60)
NOL$_{t-1}$		0.018	0.085*
		(0.80)	(1.87)
Observations	19,523	19,523	19,523
Adj. R^2	0.265	0.270	0.481
Sample	All buyers	All buyers	All buyers
Cohort × firm FE	X	X	X
Cohort × industry × year FE	X	X	X
Lagged controls		X	X

Notes: This table reports stacked-cohort difference-in-differences estimates of the effect of tax credit purchases on the GAAP ETR (Panel A) and Cash ETR (Panel B). *Entry year* is the effect in the year the credit is first recognized ($t = 0$) and *After entry* the effect in later years ($t \geq 1$). Columns (1) and (2) use the baseline control group and column (3) uses entropy balancing. Column (1) includes no controls; columns (2) and (3) add one-year-lagged log total assets, ROA, leverage, and NOL. All columns include cohort × firm and cohort × 4-digit SIC industry × year fixed effects; industry-year cells with a single observation are dropped, which reduces the number of treated firms. Cash taxes paid may include the price paid for the credits. t-statistics clustered by firm are in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table 13: Buyer-Side Responses – Sustainability Reports

Firm	Year	Excerpt from sustainability report
Tractor Supply	2024	*We also support U.S. manufacturing and lower-emission transportation fuels through the purchase of transferable federal tax credits.*
Tradeweb Markets Inc.	2024	*In 2024, Tradeweb did purchase Transferable Tax Credits from a renewable energy facility. Transferable tax credits may be considered an* ***indirect infrastructure investment as their intention is to allow indirect support for the development of renewable energy projects.***
Market Axess Holdings Inc.	2024	*In 2024, the Company continued purchasing renewable energy-related transferable tax credits under the Inflation Reduction Act (the "IRA"). The transferability market created by the IRA allows producers of renewable energy or related manufacturing components to sell the tax credits generated by their activities to corporate purchasers. Through the purchase of these credits, the Company has reduced its federal tax liability while* ***supporting innovative companies as they continue to drive the United States' transition to renewable energy.***
Nasdaq Inc.	2024	*In 2024, Nasdaq purchased transferable tax credits from two producers of renewable energy in the U.S. The transferability market allows producers of renewable energy or related manufacturing components to sell the tax credits generated by their activities to corporate purchasers. Through the purchase of these credits, Nasdaq has reduced its federal tax liability while In 2024, the Company continued purchasing renewable energy-related transferable tax credits under the Inflation Reduction Act (the "IRA"). The transferability market created by the IRA allows producers of renewable energy or related manufacturing components to sell the tax credits generated by their activities to corporate purchasers. Through the purchase of these credits, the Company has reduced its federal tax liability while* ***supporting innovative companies as they continue to drive the United States' transition to renewable energy.***
Stifel Financial Corp.	2025	*Stifel directly purchased over $100 million in renewable transfer tax credits* ***to further bolster our own commitment to the energy transition.***

Notes: The table shows excerpts from corporate sustainability in which firms disclose the purchase of transferable tax credits.

Appendix

The next pages include the following appendices:

- **Tax Credit Transfer Market** (Appendix A):
- **Seller-Side Responses** (Appendix B):

  Appendix B.1: Variable Definitions

  Appendix B.2: Additional Tables and Figures
- **Buyer-Side Responses** (Appendix C)

  Appendix C.1: Variable Definitions

  Appendix C.2: Additional Tables and Figures

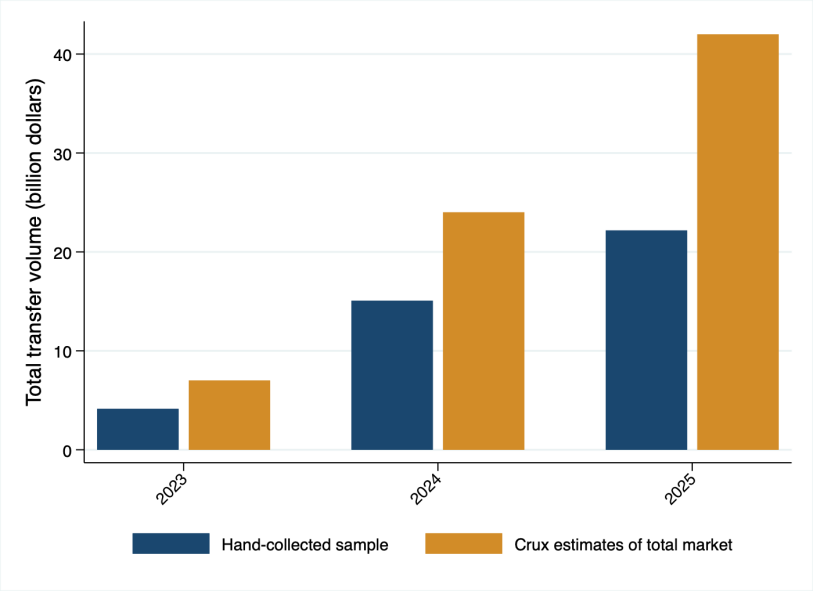
A The New Market for Tax Credits

Table A.1: Transferable Clean Energy Tax Credits

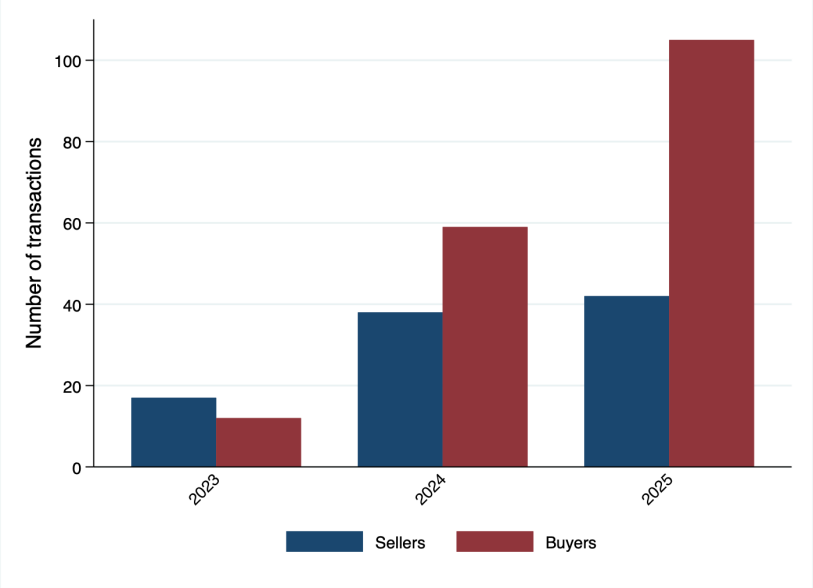
Tax Credit	Code	Technology	Credit Base	Created	First Effective
Production Tax Credit	§45	Wind	Electricity produced	1992	1992
Investment Tax Credit	§48	Solar, storage	Investment costs	1992	1992
Alternative Fuel Refueling Property Credit	§30C	EV charging, alt-fuel refueling	Investment costs	2005	2006
Carbon Capture and Sequestration Credit	§45Q	Carbon capture	CO_2 captured	2008	2008
Advanced Energy Project Credit	§48C	Clean energy production facilities	Investment costs	2009	2009
Technology-Neutral Clean Electricity Credits	§45Y/§48E	Successors to §45/§48	Investment & production	2022 (IRA)	2025
Clean Hydrogen Production Credit	§45V	Hydrogen	Clean hydrogen produced	2022 (IRA)	2023
Advanced Manufacturing Production Credit	§45X	Clean energy components	Produced goods/sales	2022 (IRA)	2023
Zero-Emission Nuclear Power Credit	§45U	Nuclear plants	Electricity produced	2022 (IRA)	2024
Clean Fuel Production Credit	§45Z	Sustainable fuels	Fuel produced	2022 (IRA)	2025

Notes: This table lists the clean energy tax credits eligible for transfer under Section 6418 of the Internal Revenue Code, introduced by the Inflation Reduction Act (IRA) of 2022, reporting for each its Code section (*Code*), the incentivized technology or activity (*Technology*), the basis on which the credit is computed (*Credit Base*), and the years it was created and became effective.

Figure A.1: Tax Credit Transfers



(a) Benchmarking hand-collected sample to Crux market estimates



(b) Number of tax credit transfers

Notes: Panel (a) compares the total transfer volume ($ billions) per year in the hand-collected sample to Crux's estimates of the total transferability market, based on its market intelligence reports (Crux 2025); missing transfer prices are imputed using the within credit-type by technology group mean. Panel (b) shows the number of tax credit transfer transactions per year, based on hand-collected transfers from firm disclosures and announcements, separately for sellers and buyers, with each transaction counted once per deal agreement. Panel (b) shows the total transfer value ($ billions) per year, separately for sellers and buyers. The sample period is 2023 to 2025.

Table A.2: Clean versus Non-clean Energy Generation

Category	Classification	EIA Source Codes
Solar	Clean	SUN
Wind	Clean	WND
Water (hydroelectric)	Clean	WAT, H2
Geothermal	Clean	GEO
Biomass	Clean	LFG, OBG, AB, MSW, OBS, OBL, BLQ, WDL, SLW, WH, WDS
Battery storage	Clean	MWH
Nuclear	Clean	NUC
Coal	Non-clean	RC, LIG, PT, WC, SUB, BIT, SGC
Gas	Non-clean	NG, OG, BFG, PG
Petroleum	Non-clean	SGP, JF, KER, DFO, RFO, WO, PC

Notes: This table reports how generators are classified as clean or non-clean throughout the paper, based on each generator's primary energy source in EIA Form 860 (*EnergySource1*): clean if solar, wind, hydroelectric, geothermal, biomass, battery storage, or nuclear, and non-clean if coal, gas, or petroleum.

B Seller-Side Responses

B.1 Variable Definition Seller-Side Responses

Table B.1: Seller-Side Responses – Variable Definitions

Variable	Definition	Data source
$\text{CapEx}_{i,t}$	Capital expenditures for firm i in year t, in \$ millions.	Compustat (`capx`)
$\text{CapEx investment}_{i,t}$	CapEx divided by firm i's total assets at the end of year $t-1$.	Compustat ($\texttt{capx/at}_{t-1}$)
$\text{PPE}_{i,t}$	Gross property, plant, and equipment for firm i in year t, in \$ millions.	Compustat (`ppegt`)
$\text{PPE investment}_{i,t}$	PPE divided by firm i's total assets at the end of year $t-1$.	Compustat ($\texttt{ppegt/at}_{t-1}$)
$\text{ROA}_{i,t}$	Pretax income divided by firm i's total assets at the end of year $t-1$.	Compustat ($\texttt{pi/at}_{t-1}$)
$\text{Total assets}_{i,t}$	Total assets for firm i in year t, in \$ millions.	Compustat (`at`)
$\text{Pretax income / assets}_{i,t}$	Pretax income divided by total assets for firm i in year t.	Compustat (`pi/at`)
$\text{Leverage}_{i,t}$	Long-term debt (plus the current portion of long-term debt, where available) divided by total assets for firm i in year t.	Compustat (`dltt[+dlc]/at`)
$\text{Number of generators}_{i,t}$	Number of distinct electricity generating units operated by firm i in year t.	EIA Form 860
$\text{Clean / Non-clean generators}_{i,t}$	Number of generators using clean (solar, hydroelectric, wind) / non-clean generating technologies operated by firm i in year t.	EIA Form 860
$\text{Total capacity}_{i,t}$	Total nameplate capacity across all generators operated by firm i in year t, in megawatts (MW).	EIA Form 860
$\text{Clean / Non-clean capacity}_{i,t}$	Total nameplate capacity, in MW, across clean / non-clean generators operated by firm i in year t.	EIA Form 860
$\text{Net generation}_{i,t}$	Total net electricity generation across all generators operated by firm i in year t, in gigawatt-hours (GWh).	EIA Form 923
$\text{Pct. generation clean}_{i,t}$	Share of net generation from clean generating technologies (solar, hydroelectric, wind) out of total net generation for firm i in year t.	EIA Forms 860, 923
$\text{Capacity factor}_{i,t}$	Net generation divided by nameplate capacity $\times$ 8,760 hours for firm i in year t.	EIA Forms 860, 923
$\text{Capacity factor of projects}_{j,t}$	Net generation divided by nameplate capacity $\times$ 8,760 hours for solar or wind project j, an EIA Form 860 generator, in year t. For new projects, it is measured in the first full calendar year after commissioning, and for existing projects in each full operating year.	EIA Forms 860, 923
$\text{Average generation}_{i,t}$	Net generation divided by the number of generators, excluding battery storage units, operated by firm i in year t, in gigawatt-hours (GWh).	EIA Forms 860, 923

B.2 Additional Tables and Figures Seller-Side Responses

Table B.2: Seller-Side Responses – Summary Statistics for the Firm-Cohort-Year Panel

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Stacked-cohort sample				2023 sample			
	N	Mean	SD	P50	N	Mean	SD	P50
Panel A: *Public electric utility firm-cohort-year panel*								
Dependent variables:								
CapEx	2,473	2,071.83	3,001.12	929.00	1,081	1,982.68	2,940.21	876.00
CapEx investment	2,473	0.06	0.04	0.06	1,081	0.06	0.04	0.06
Log(CapEx + 1)	2,473	6.63	1.76	6.84	1,081	6.55	1.81	6.78
ROA	2,473	0.05	0.08	0.04	1,081	0.04	0.09	0.04
Generators	2,473	32.04	53.14	8.00	1,081	31.59	53.88	8.00
Log(Generators + 1)	2,473	2.51	1.39	2.20	1,081	2.48	1.39	2.20
Total capacity	2,473	3,342.44	6,828.51	275.00	1,081	3,293.66	6,819.53	252.00
Log(Total capacity + 1)	2,473	5.68	2.72	5.62	1,081	5.61	2.75	5.53
Capacity factor	2,473	0.40	0.21	0.38	1,081	0.40	0.22	0.38
Control variables:								
Log(Total assets$_{t-1}$)	2,473	9.78	1.52	9.70	1,081	9.74	1.53	9.66
ROA$_{t-1}$	2,473	0.05	0.08	0.04	1,081	0.05	0.09	0.04
Leverage$_{t-1}$	2,473	0.36	0.14	0.35	1,081	0.36	0.15	0.35
Log(Net generation$_{t-1}$ + 1)	2,473	13.38	3.67	13.56	1,081	13.30	3.71	13.51
Pct. clean generation$_{t-1}$	2,473	0.37	0.43	0.09	1,081	0.38	0.43	0.10
Cross-sectional variables:								
Clean generators	2,473	17.31	37.53	3.00	1,081	17.19	38.43	3.00
Log(Clean generators + 1)	2,473	1.62	1.52	1.39	1,081	1.61	1.52	1.39
Non-clean generators	2,473	14.72	25.55	3.00	1,081	14.38	25.28	3.00
Log(Non-clean generators + 1)	2,473	1.73	1.44	1.39	1,081	1.70	1.44	1.39
NOL/assets	933	0.16	0.34	0.05	422	0.18	0.39	0.06
Pretax income	2,473	2,929.24	6,909.36	647.00	1,081	2,598.33	6,345.59	588.25
Ever tax equity	2,473	0.04	0.19	0.00	1,081	0.03	0.18	0.00
Total assets	2,470	53,743.07	105,189.09	17,344.00	1,078	53,518.10	110,797.85	17,221.00
HP index	2,473	-4.36	0.39	-4.64	1,081	-4.34	0.40	-4.64
OCF	2,470	4,861.23	12,910.33	1,127.00	1,078	4,802.93	13,259.02	986.00

Table B.2: Seller-Side Responses – Summary Statistics for the Firm-Cohort-Year Panel (continued)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Stacked-cohort sample				2023 sample			
	N	Mean	SD	P50	N	Mean	SD	P50
Panel B: *Public and private electric utility firm-cohort-year panel*								
Dependent variables:								
Generators	58,344	5.14	17.73	1.00	21,512	5.26	17.55	1.00
Log(Generators + 1)	58,344	1.20	0.79	0.69	21,512	1.22	0.80	0.69
Total capacity	58,344	231.41	754.59	16.00	21,512	247.53	798.17	17.10
Log(Total capacity + 1)	58,344	3.23	2.04	2.83	21,512	3.27	2.06	2.90
Capacity factor	58,344	0.31	0.22	0.24	21,512	0.31	0.22	0.25
Control variables:								
Log(Net generation$_{t-1}$ + 1)	58,344	10.31	3.21	10.34	21,512	10.39	3.19	10.42
Pct. clean generation$_{t-1}$	58,344	0.69	0.46	1.00	21,512	0.68	0.46	1.00
Cross-sectional variables:								
Clean generators	58,344	3.29	14.15	1.00	21,512	3.36	14.24	1.00
Log(Clean generators + 1)	58,344	0.81	0.80	0.69	21,512	0.81	0.81	0.69
Non-clean generators	58,344	1.84	8.33	0.00	21,512	1.90	7.48	0.00
Log(Non-clean generators + 1)	58,344	0.45	0.81	0.00	21,512	0.47	0.82	0.00

Notes: This table reports summary statistics for the stacked-cohort panel, in which each treated cohort is paired with all never-treated control firms and control firm-years are repeated once per cohort stack. Panel A reports Sample 1 (public electric utility firm-cohort-year panel) and Panel B reports Sample 2 (public and private electric utility firm-cohort-year panel). Columns (1) through (4) report the full stacked-cohort sample and columns (5) through (8) restrict it to the 2023 cohort. Variables are reported in the form used in the analysis, including level, log, and capacity outcomes, the one-year-lagged form of each control, and the clean/non-clean breakdown of generators. Panel A's cross-sectional variables are reported in the same form used to construct the splits in Table 5: ROA, NOL/assets, and pretax income are each firm's pre-treatment mean; total assets, the HP index, and OCF are measured in event year $t-1$; ever tax equity is a time-invariant indicator. Generation is in GWh, capacity in MW, and financial variables in millions of dollars. All variables are defined in Appendix B.1.

Table B.3: Seller-Side Responses – Pre-treatment Comparison of Treatment and Control Firms

	Stacked-cohort sample					2023 sample				
	Mean			Diff. in Means		Mean			Diff. in Means	
	Treatment	Baseline control	Entr. balanced control	Unweighted	Weighted	Treatment	Baseline control	Entr. balanced control	Unweighted	Weighted
Panel A: *Public electric utility firm-cohort-year panel (pre-treatment)*										
Log(Total assets)	9.62	9.79	9.62	-0.17	0.00	9.75	9.71	9.75	0.03	-0.00
Pretax income/assets	-0.00	0.05	-0.00	-0.05***	-0.00	0.01	0.05	0.01	-0.04**	-0.00
Leverage	0.36	0.32	0.36	0.03	0.00	0.34	0.32	0.34	0.02	-0.00
Log(total capacity)	7.67	5.55	7.67	2.13***	0.00	8.05	5.35	8.05	2.70***	0.00
Pct. clean generation	0.47	0.38	0.47	0.10	0.00	0.35	0.40	0.35	-0.05	-0.00
N	21	408	408			11	157	157		
Panel B: *Public and private electric utility firm-cohort-year panel (pre-treatment)*										
Log(Net generation)	13.80	10.46	13.79	3.34***	0.01	15.06	10.51	15.05	4.55***	0.01
Pct. clean generation	0.58	0.70	0.58	-0.12*	-0.00	0.49	0.69	0.49	-0.20*	0.00
Log(total capacity)	6.34	3.19	6.34	3.16***	0.01	7.01	3.20	7.00	3.81***	0.00
N	44	10,239	10,239			19	3,250	3,250		

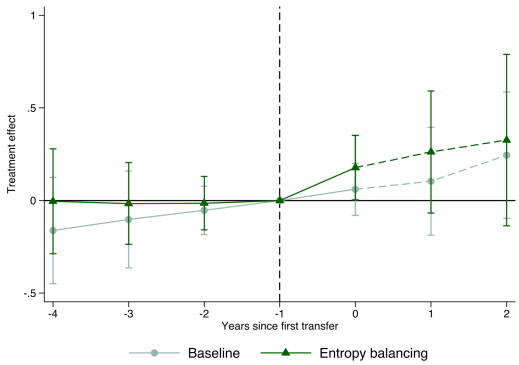
Notes: This table compares treated and control firms before treatment on the entropy balancing target variables for each sample: Panel A reports Sample 1 (public electric utility firm-cohort-year panel) and Panel B reports Sample 2 (public and private electric utility firm-cohort-year panel). The first five columns use the full stacked-cohort sample and the last five restrict it to the 2023 cohort. Each row is a firm-stack's own pre-treatment mean; the treatment group is unweighted throughout, and N reports the number of firm-stack observations in each group. *Diff. in Means, Unweighted* reports the treated-minus-control difference from a two-sample t-test against the baseline control group; *Weighted* reports the treatment coefficient from a regression against the entropy-balancing-weighted control group, with robust standard errors. All variables are defined in Appendix B.1. *** p<0.01, ** p <0.05, * p<0.1.

Table B.4: Seller-Side Responses – Robustness for Effect of Tax Credit Transfer on Investment

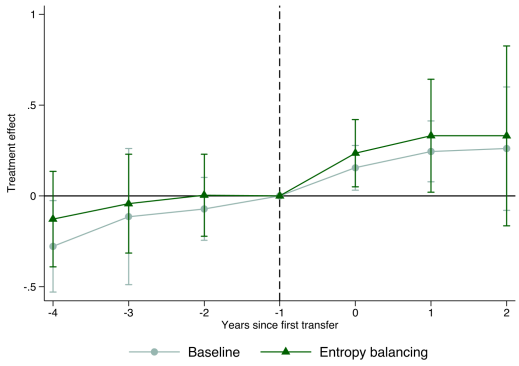
	(1)	(2)	(3)	(4)
	Entropy balanced			
	CapEx investment		Log(CapEx + 1)	
Treat × **Post**	0.012*	0.021**	0.236**	0.324**
	(1.86)	(2.48)	(2.01)	(2.48)
Log(Total assets)$_{t-1}$			1.409***	0.688***
			(4.53)	(4.03)
Pretax income / assets$_{t-1}$			0.185	0.150
			(0.49)	(0.43)
Leverage$_{t-1}$			-0.808	-0.306
			(-1.27)	(-0.95)
Observations	2,473	1,081	2,473	1,081
Adj. R^2	0.679	0.695	0.967	0.972
Sample	Public	2023	Public	2023
Cohort × firm FE	X		X	
Cohort × year FE	X		X	
Firm FE		X		X
Year FE		X		X
Lagged controls			X	X
Pre-treatment controls	X	X		

Notes: This table reports robustness checks for the main results on capital expenditures. All specifications here use entropy balancing. Columns (1)-(2) report the investment-ratio (level) outcome under the pre-treatment mean-x-year control specification: column (1) on Sample 1 (public electric utility firm-cohort-year panel) and column (2) restricted to the 2023 cohort, the only cohort with a full post-treatment window. Columns (3)-(4) instead use the alternative log(1 + y) transformation of the outcome with one-year-lagged controls: column (3) on Sample 1; column (4) restricted to the 2023 cohort. Both 2023-cohort columns use firm and year fixed effects in place of the cohort-interacted terms. The log(1 + y), no-controls specification is omitted. All variables are defined in Appendix B.1. t-statistics based on standard errors clustered at the firm level are in parentheses. *** p<0.01, ** p <0.05, * p<0.1.

Figure B.1: Seller-Side Responses – Effect of Tax Credit Transfer on Investment using Logs, Event Studies



(a) Log(CapEx + 1)



(b) Log(CapEx + 1), 2023 cohort

Notes: This figure plots event-study estimates of the effect of tax credit transfers on investment, using the alternative log(1 + y) transformation of CapEx, obtained by replacing the *Post* indicator with separate relative-year indicators. Panel (a) uses Sample 1, the public electric utility firm-cohort-year panel, with cohort $\times$ firm and cohort $\times$ year fixed effects; Panel (b) restricts Sample 1 to the 2023 cohort, the only cohort with a full post-treatment window, with firm and year fixed effects in place of the cohort-interacted terms. Each regression includes one-year-lagged control variables. Each panel presents the baseline control group (turquoise circles) and entropy balancing (dark green triangles). In Panel (a), the series are dashed at event years $t+1$ and $t+2$ because the 2024 and 2025 cohorts are only partially represented or not represented at these horizons. Capped vertical lines indicate 95% confidence intervals; standard errors are clustered at the firm level.

Table B.5: Seller-Side Responses – Robustness for Effect of Tax Credit Transfer on Generator Investment

	(1)	(2)	(3)	(4)	(5)	(6)
	Entropy balanced					
Panel A:	*Log(Generators + 1)*			*Generators, PPML*		
Treat × Post	0.106***	0.105**	0.043	0.085**	0.098**	0.113**
	(2.67)	(2.09)	(1.55)	(2.16)	(2.00)	(2.07)
Log(Net gen + 1)$_{t-1}$				0.150***	0.174***	0.124**
				(4.22)	(3.43)	(2.48)
Pct clean gen$_{t-1}$				0.161	0.317	-0.258
				(0.73)	(1.33)	(-1.07)
Observations	2,473	1,081	58,344	2,473	1,081	58,344
Adj. R^2 / Pseudo R^2	0.979	0.971	0.989	0.895	0.891	0.916
Panel B:	*Log(Clean generators + 1)*			*Clean generators, PPML*		
Treat × Post	0.142**	0.156*	0.078**	0.131**	0.163**	0.164***
	(2.21)	(1.68)	(2.00)	(2.15)	(2.13)	(2.79)
Log(Net gen + 1)$_{t-1}$				0.095***	0.120**	0.196***
				(2.97)	(2.57)	(4.85)
Pct clean gen$_{t-1}$				0.461**	0.720***	-0.470
				(2.41)	(3.03)	(-1.02)
Observations	2,473	1,081	58,344	1,845	806	44,156
Adj. R^2 / Pseudo R^2	0.986	0.981	0.987	0.898	0.888	0.915
Panel C:	*Log(Non-clean generators + 1)*			*Non-clean generators, PPML*		
Treat × Post	0.016	0.044	0.004	0.006	0.011	-0.023
	(0.32)	(0.81)	(0.32)	(0.17)	(0.28)	(-1.03)
Log(Net gen + 1)$_{t-1}$				0.222***	0.235***	0.009**
				(4.15)	(3.50)	(2.48)
Pct clean gen$_{t-1}$				-0.283	-0.223	-0.098
				(-0.75)	(-0.56)	(-0.79)
Observations	2,473	1,081	58,344	1,866	804	17,698
Adj. R^2 / Pseudo R^2	0.977	0.970	0.998	0.841	0.829	0.832
Sample	Public	2023	+ Private	Public	2023	+ Private
Cohort × firm FE	X		X	X		X
Cohort × year FE	X		X	X		X
Firm FE		X			X	
Year FE		X			X	
Pre-treatment controls	X	X	X			
Lagged controls				X	X	X

Notes: This table reports robustness checks for the effect of tax credit transfers on the number of generators: overall (Panel A), clean (Panel B), and non-clean (Panel C). All columns use entropy balancing. Columns (1) through (3) use log(count + 1) with pre-treatment controls (pre-treatment mean log net generation and clean generation share, each interacted with year), whereas columns (4) through (6) use the generator count in a PPML estimation with one-year-lagged controls; PPML drops fixed-effect groups without variation in the outcome. Within each set, the first column uses Sample 1, the public electric utility firm-cohort-year panel; the second restricts Sample 1 to the 2023 cohort, with firm and year fixed effects in place of the cohort-interacted terms; the third uses Sample 2, the public and private electric utility firm-cohort-year panel. All variables are defined in Appendix B.1. t-statistics based on standard errors clustered at the firm level are in parentheses. *** p<0.01, ** p <0.05, * p<0.1.

Table B.6: Seller-Side Responses – Cross-Sectional Tests by Prior Clean Energy Exposure

	(1)	(2)	(3)	(4)
	Entropy balanced			
	Pre-clean gen		Pre technologies	
	Low	High	Few	Multiple
CapEx investment				
Treat × **Post**	0.022**	-0.008	0.015	0.010
	(2.56)	(-0.89)	(1.65)	(1.26)
p-value, difference	0.033**		0.941	
Observations	1,569	904	1,142	1,331
Adj. R^2	0.659	0.640	0.725	0.569
Log(Clean generators + 1)				
Treat × **Post**	0.131	0.104*	0.149**	0.126
	(1.46)	(1.75)	(2.28)	(1.59)
p-value, difference	0.323		0.780	
Observations	1,569	904	1,142	1,331
Adj. R^2	0.985	0.988	0.989	0.980
Sample	Public	Public	Public	Public
Cohort × firm FE	X	X	X	X
Cohort × year FE	X	X	X	X
Lagged controls	X	X	X	X

Notes: This table reports the Treat × Post coefficient from separate subsample regressions examining which firms expand clean investment after the first tax credit transfer, using Sample 1 (public electric utility firm-cohort-year panel). The dependent variables are CapEx scaled by one-year-lagged total assets and log(Clean generators + 1). All columns use entropy balancing and one-year-lagged controls. The splits use each firm's pre-treatment clean generation share (Pre-clean gen, split at 0.5) and the number of distinct clean technologies deployed (Pre technologies, split at the sample median); both are each firm's pre-treatment mean, and the median cutoff is computed once on the baseline (non-entropy-balanced) sample and applied identically to the entropy-balanced sample. Each column omits whichever lagged control mechanically overlaps with that column's own split variable. The row labeled "p-value, difference" reports the p-value on the Treat × Post × group interaction from a pooled regression estimated on the two subsamples together, testing whether the Treat × Post coefficient differs between the two groups; the group indicator itself is absorbed by the cohort × firm fixed effects. All variables are defined in Appendix B.1. Standard errors are clustered at the firm level; t-statistics are reported in parentheses beneath the coefficients. Significance levels are based on two-sided t-tests: * p<0.10, ** p<0.05, *** p<0.01.

C Buyer-Side Responses

C.1 Variable Definition Buyer-Side Responses

Table C.1: Buyer-Side Responses – Variable Definitions

Variable	Definition	Data source
Dependent variables:		
Tax credit purchase$_{i,t}$	Indicator equal to one if firm i purchased transferable tax credits in year t, and zero otherwise.	Hand-collected
GAAP ETR$_{i,t}$	Total income tax expense divided by pretax income net of special items for firm i in year t, winsorized at 0 and 1.	Compustat (txt/(pi-spi))
Cash ETR$_{i,t}$	Income taxes paid in cash divided by pretax income net of special items for firm i in year t, winsorized at 0 and 1.	Compustat (txpd/(pi-spi))
Control variables:		
Log(Total assets)$_{i,t-1}$	Natural logarithm of total assets for firm i measured in year $t-1$ (one-year lag).	Compustat (at)
ROA$_{i,t-1}$	Net income divided by total assets for firm i measured in year $t-1$ (one-year lag).	Compustat (ni/at)
Leverage$_{i,t-1}$	Long-term debt divided by total assets for firm i measured in year $t-1$ (one-year lag).	Compustat (dltt/at)
NOL$_{i,t-1}$	Tax loss carryforward divided by total assets (missing carryforward treated as zero) for firm i measured in year $t-1$ (one-year lag).	Compustat (tlcf/at)
Cross-sectional variables:		
Log(Total assets)$_{i,t}$	Natural logarithm of total assets for firm i in year t.	Compustat (at)
Log(Market cap)$_{i,t}$	Natural logarithm of market capitalization for firm i in year t, where market capitalization is common shares outstanding times the fiscal-year-end closing share price, in $ millions; shares and price winsorized at the 1st/99th percentile.	Compustat (csho×prcc_f)
Firm age$_{i,t}$	Number of years firm i has appeared on Compustat, winsorized at the 1st/99th percentile.	Compustat (age)
Cash / assets$_{i,t}$	Cash and short-term investments divided by total assets for firm i in year t.	Compustat (che/at)
ROA$_{i,t}$	Net income divided by total assets for firm i in year t.	Compustat (ni/at)
Leverage$_{i,t}$	Long-term debt divided by total assets for firm i in year t.	Compustat (dltt/at)
Pretax income / assets$_{i,t}$	Pretax income divided by total assets for firm i in year t.	Compustat (pi/at)
Cash taxes / assets$_{i,t}$	Income taxes paid divided by total assets for firm i in year t.	Compustat (txpd/at)
Persistent profitability$_i$	Number of years from 2019 to 2022 in which firm i reports positive pretax income (0–4).	Compustat (pi)

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Table C.1 continued from previous page

Variable	Definition	Data source
$\text{NOL}_{i,t}$	Tax loss carryforward divided by total assets for firm i in year t (missing carryforward treated as zero).	Compustat (`tlcf/at`)
Tax equity, ever_i	Indicator equal to one if firm i ever invested in tax equity over the sample period, and zero otherwise.	Hand-collected
Book-tax $\text{difference}_{i,t}$	Pretax book income less estimated taxable income, divided by total assets for firm i in year t. Estimated taxable income is total income tax expense divided by the statutory federal tax rate (35% before 2018, 21% from 2018).	Compustat (`(pi-txt/`τ`)/at`)
Foreign $\text{operations}_{i,t}$	Indicator equal to one if firm i reports positive foreign income taxes in year t (missing treated as zero), and zero otherwise.	Compustat (`txfo`)
ESG $\text{score}_{i,t}$	Combined ESG score for firm i in year t, 0–100 scale, winsorized at the 1st/99th percentile.	LSEG (Refinitiv)
Emission red. $\text{target}_{i,t}$	Indicator equal to one if firm i has set an emissions reduction target in year t, and zero otherwise.	LSEG (Refinitiv)

C.2 Additional Tables and Figures Buyer-Side Responses

Table C.2: Tax Equity Investment and Tax Credit Purchases by Year

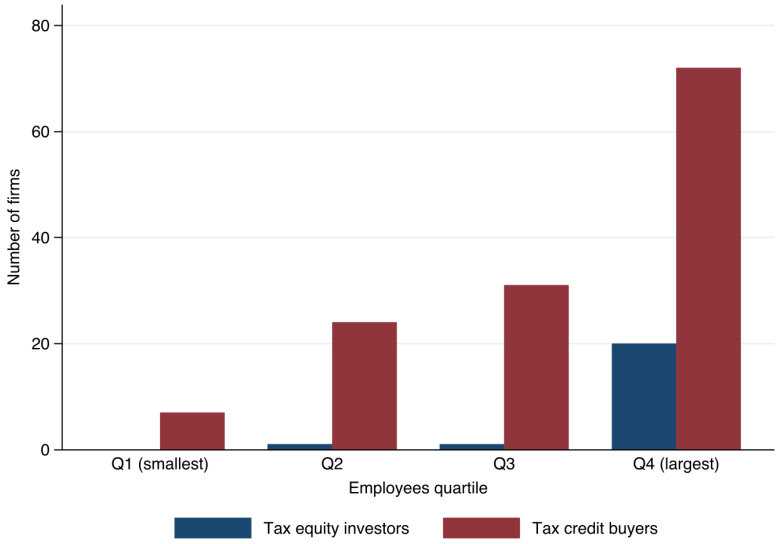
Year	Tax Equity Investment	Tax Credit Purchase
2015	6	0
2016	11	0
2017	9	0
2018	7	0
2019	6	0
2020	12	0
2021	7	0
2022	5	0
2023	9	14
2024	6	66
2025	4	126
Total	82	206

Notes: The table reports the number of firms with tax equity investments and tax credit purchases per year. A firm is counted in a given year if it invests in tax equity or purchases a tax credit in that year. The sample period is 2015 to 2025.

Figure C.1: Market Participation by Firm Size: Size Quartiles



(a) Total assets



(b) Employees

Notes: The figure shows the number of tax equity investors (navy) and tax credit buyers (maroon) by firm-size quartile, based on the firm's pre-treatment (2019–2022) mean size. Panel (a) uses total assets and Panel (b) uses number of employees to construct quartiles of firm size.

Table C.3: Buyer-Side Responses – Summary Statistics for the Firm-Cohort-Year Panel

	N	Mean	SD	P25	P50	P75
Firm-cohort-year panel (Sample 4)						
Dependent variables:						
Tax credit purchase	19,523	0.01	0.10	0.00	0.00	0.00
GAAP ETR	19,523	0.21	0.14	0.15	0.21	0.25
Cash ETR	19,523	0.20	0.17	0.09	0.18	0.26
Control variables:						
Log(Total assets$_{t-1}$)	19,523	8.73	2.12	7.38	8.59	10.02
ROA$_{t-1}$	19,523	0.04	0.08	0.01	0.02	0.06
Leverage$_{t-1}$	19,523	0.19	0.21	0.02	0.11	0.30
NOL$_{t-1}$	19,523	0.04	0.19	0.00	0.00	0.01
Cross-sectional variables:						
Log(Total assets)	19,523	8.81	2.10	7.46	8.67	10.09
Log(Market cap)	18,426	7.81	2.22	6.16	7.83	9.37
Firm age	19,523	26.45	17.74	13.00	23.00	35.00
Cash / assets	19,523	0.11	0.13	0.03	0.07	0.14
ROA	19,523	0.04	0.09	0.01	0.02	0.06
Leverage	19,519	0.19	0.21	0.02	0.11	0.31
Pretax income / assets	19,523	0.05	0.10	0.01	0.03	0.08
Cash taxes / assets	19,523	0.01	0.02	0.00	0.00	0.02
Persistent profitability	19,523	3.78	0.42	4.00	4.00	4.00
NOL	19,523	0.04	0.18	0.00	0.00	0.01
Tax equity, ever	19,523	0.02	0.15	0.00	0.00	0.00
Book-tax difference	19,523	-0.01	0.55	-0.01	-0.00	0.00
Foreign operations	19,523	0.33	0.47	0.00	0.00	1.00
ESG score	12,689	44.93	18.82	30.50	43.05	59.60
Emission red. target	14,241	0.16	0.37	0.00	0.00	0.00

Notes: This table reports summary statistics for the stacked-cohort panel (Sample 4), in which each treated cohort, defined by the audited entry year, is paired with all never-treated control firms and control firm-years are repeated once per cohort. Control variables are measured in year $t-1$. All variables are defined in Appendix C.1.

Table C.4: Buyer-Side Responses – Tax Benefit of Purchasing Tax Credit Sample

	Observations	Firms
Hand-collected tax credit buyer disclosures 2023–2025		
All buyer disclosures (Transactions_Purchases)	182	116
Less no Compustat pretax income for the fiscal year	(37)	(25)
Disclosures with Compustat pretax income	145	91
Less pretax income not positive	(6)	(3)
Less no benefit measure in filing	(96)	(56)
Tax benefit sample	**43**	**32**

Notes: This table reports the sample used to estimate the tax benefit of purchasing tax credits. I match hand-collected buyer disclosures to Compustat pretax income by CIK and fiscal period end and verify every transaction against the 10-K. The sample keeps transactions with positive pretax income and at least one of three measures: the credit face value less the price paid, the tax benefit reported as a reduction of tax expense, or the effect on the effective tax rate in the rate reconciliation. A transaction can enter more than one measure.

Table C.5: Buyer-Side Responses – Accounting for Purchased Tax Credits

	Credit applied to this year's tax	Credit applied later
Panel A: Recording a tax credit purchase		
1. Tax expense accrual in year t	Dr Income tax expense 210 Cr Income taxes payable 210	Dr Income tax expense 210 Cr Income taxes payable 210
2. Purchase of the credit (face value 100, price 93)	Dr Income taxes payable 100 Cr Cash 93 Cr Income tax expense 7	Dr Income tax receivable 100 Cr Cash 93 Cr Income tax expense 7
3. Payment to the IRS in year t	Dr Income taxes payable 110 Cr Cash 110	Dr Income taxes payable 210 Cr Cash 210
4. Year $t+1$	—	Dr Cash 100 (refund or lower payments) Cr Income tax receivable 100
Panel B: Resulting taxes paid and effective tax rates		
Paid to the IRS$_t$	$210-100=110$	210
Paid to the seller$_t$	93	93
Cash taxes paid$_t$	**203**	**303**
GAAP income tax expense$_t$	$210-7=203$	$210-7=203$
Pretax income$_t$	1,000	1,000
GAAP ETR$_t$	$203/1{,}000=20.3\%$	$203/1{,}000=20.3\%$
Cash ETR$_t$	$203/1{,}000=20.3\%$	$303/1{,}000=30.3\%$

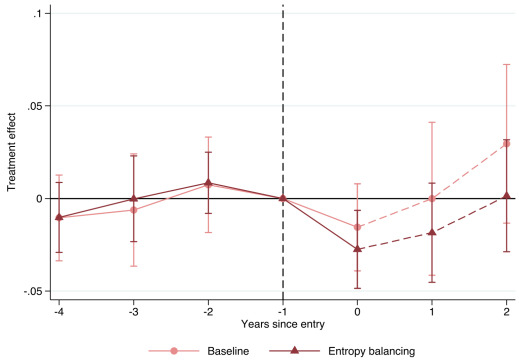
Notes: This table illustrates how a firm records a purchased tax credit with a face value of 100 and a price of 93 when pretax income is 1,000 and the tax before credits is 210. Dr = debit and Cr = credit. The discount of 7 lowers GAAP income tax expense at purchase, whereas cash taxes paid include the price paid to the seller. If the credit is applied after the year's estimated payments, against the prior year's liability, or as a carryback, it is an income tax receivable, so Cash ETR$_t$ is higher and Cash ETR$_{t+1}$ is lower.

Table C.6: Buyer-Side Responses – Pre-Treatment Comparison of Treatment and Control Firms

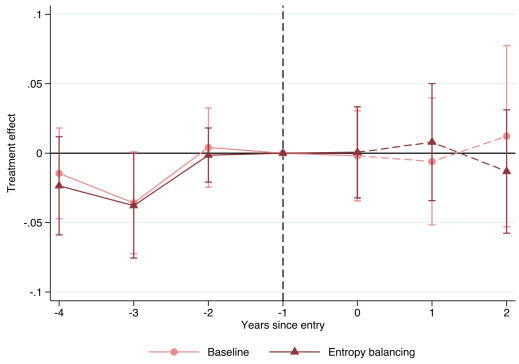
	Treatment		Baseline control		Entr. balanced control		Diff. in Means	
	N	Mean	N	Mean	N	Mean	Unweighted	Weighted
All buyers firm-cohort-year panel (pre-treatment)								
Log(Total assets)	119	9.46	3,245	8.71	3,245	9.46	0.75***	0.00
ROA	119	0.06	3,245	0.04	3,245	0.06	0.02***	0.00
Leverage	119	0.27	3,245	0.19	3,245	0.27	0.08***	0.00
NOL / assets	119	0.03	3,245	0.04	3,245	0.03	-0.02	-0.00
Pretax income / assets	119	0.08	3,245	0.05	3,245	0.08	0.02***	0.00

Notes: This table compares treated and control firms before entry on the entropy balancing target variables for all buyers. Each row is a firm-cohort pair's own pre-entry mean, and the treatment group is unweighted throughout. Diff. in Means, Unweighted reports the treated-minus-control difference from a two-sample t-test against the baseline control group; Weighted reports the treatment coefficient from a regression against the entropy-balanced control group, with robust standard errors. All variables are defined in Appendix C.1. *** p<0.01, ** p<0.05, * p<0.1.

Figure C.2: Buyer-Side Responses – Event Studies of the Effect of Tax Credit Purchase on ETR



(a) GAAP ETR



(b) Cash ETR

Notes: This figure plots event-study estimates of the effect of tax credit purchases on the GAAP ETR (Panel (a)) and the Cash ETR (Panel (b)) for all buyers. Each regression includes cohort $\times$ firm and cohort $\times$ 4-digit industry $\times$ year fixed effects and one-year-lagged control variables. Each panel presents estimates using the baseline control group (light red circles) and entropy balancing (dark red triangles). The year before entry ($t = -1$) is the reference, and the series are dashed from the entry year onward. Capped vertical lines indicate 95% confidence intervals. Standard errors are clustered at the firm level.